Model Architecture: Qwen3.5-9B

Generated by sg_modelload

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Contents

1 Model Overview **2**

2 Mathematical Representation **2**
2.1 The model as a composition of layers . . . 2
2.2 The layers, top to bottom . . . 3
2.3 Neurons and attention heads . . . 5
2.4 Architecture and parameters . . . 5
2.5 Output representations . . . 5
2.6 Intermediate states and sub-network decomposition . . . 6
2.7 Local linear representation . . . 6
2.8 Summary of notation . . . 6

3 Forward Pass in Pseudocode **7**

4 Compact Architecture (repeated layers written once) **8**
4.1 Modules outside the repeated stacks . . . 8
4.2 Language-model decoder: `model.language_model.layers` . . . 9
4.3 Vision encoder: `model.visual.blocks` . . . 9
4.4 Checkpoint-only modules . . . 10

5 What Each Layer Does **10**

6 Parameter Count **12**

A Full Layer-by-Layer Table **12**

B Per-Layer Parameter Count **31**

1 Model Overview

Model name	Qwen3.5-9B
Transformers class	`Qwen3_5ForConditionalGeneration`
Path	`/models/sayak_da_work/Qwen3.5-9B`
Total parameters	9,409,813,744 (9.41B)
Checkpoint-only tensors (not in total)	243,290,624
Weight dtype	bfloat16
Hidden size d	4,096
Decoder layers L	32
Query heads H	16
Key/value heads H_{kv}	4
Head dim d_h	256
MLP width d_{ff}	12,288
DeltaNet q/k heads	16
DeltaNet v heads	32
DeltaNet q/k head dim	128
DeltaNet v head dim	128
DeltaNet conv kernel	4
Vocabulary V	248,320
Max context length	262,144
RMSNorm ϵ	1e-06
Tied input/output embeddings	False
Activation	SiLU
RoPE base θ	10,000,000
Vision encoder: layers	27
hidden size / heads	1152 / 16

How to read this document. Section 2 treats the model mathematically, top-down: first the whole network as one composition $\Psi \circ \mathcal{B}^{(L)} \circ \cdots \circ \mathcal{B}^{(1)} \circ E$ of layers, then a definition of every layer down to single neurons and attention heads, then parameters, outputs, intermediate states and the local linear response. Section 3 gives readable pseudocode for the forward pass and for every distinct block type. Section 4 lists the weights of each block type *once*, with how many times it repeats. Section 5 explains in plain language what every kind of layer does. The original layer-by-layer table with every repetition written out is in Appendix A.

Basic notation

B, T	batch size, sequence length (tokens)
d	hidden size (4096)
V	vocabulary size (248,320)
H, H_{kv}, d_h	query heads, key/value heads, size of one head
W_*, b_*	learned weight matrix / bias of the named projection
$\odot$, σ	element-wise product, logistic sigmoid
$f^{\circ n}$	f applied n times in a row ($f \circ f \circ \cdots \circ f$)
$g \circ f$	first f, then g (composition reads right to left)
RMSNorm(x)	$\gamma \odot x / \sqrt{\text{mean}(x^2) + \epsilon}$ (learned scale γ)
LN(x)	LayerNorm: $\gamma \odot (x - \mu)/\sigma + \beta$
RoPE	rotary position embedding (rotates query/key pairs by an angle set by position)

2 Mathematical Representation

We describe the model *top-down*. First the whole network is written as one composition of functions (Section 2.1). Then every function that appears in that composition is defined (Section 2.2), down to individual neurons and attention heads. The remaining subsections describe the parameters, the outputs, the intermediate states, and the local linear behaviour.

2.1 The model as a composition of layers

Definition 2.1 (Qwen3.5-9B as an L-layer decoder)**.** The model is a function $f_\theta : \mathcal{X} \to \mathcal{Y}$ with parameters θ. It maps a sequence of T token ids $x \in \mathcal{X} = \{1, \ldots, V\}^T$ to one logit vector per position, $y \in \mathcal{Y} = \mathbb{R}^{T \times V}$

($V = 248{,}320$). It is a composition of an embedding E_{mm}, $L = 32$ decoder blocks $\mathcal{B}^{(\ell)}$ and an output head Ψ:

$$f_\theta \;=\; \Psi \circ \mathcal{B}^{(L)} \circ \mathcal{B}^{(L-1)} \circ \cdots \circ \mathcal{B}^{(2)} \circ \mathcal{B}^{(1)} \circ E_{\text{mm}} \tag{1}$$

Every block maps $\mathbb{R}^{T\times d} \to \mathbb{R}^{T\times d}$ ($d = 4096$), so the blocks can be stacked in any number. Block ℓ has its own parameters θ_ℓ and is stored in the checkpoint as `model.language_model.layers.`$(\ell - 1)$.

Definition 2.2 (Block types)**.** There are 2 block types (same function form within a type, different weights). A type map $c : \{1, \ldots, L\} \to \{\text{L}, \text{F}\}$ says which type layer ℓ is: $\mathcal{B}^{(\ell)} = \mathcal{B}_{c(\ell)}(\,\cdot\,; \theta_\ell)$. Here $c = ((\text{L}^3\,\text{F})^8)$, read left to right from layer 1. Written out, with the type as a subscript:

$$f_\theta = \Psi \circ \mathcal{B}_{\text{F}}^{(32)} \circ \cdots \circ \mathcal{B}_{\text{F}}^{(4)} \circ \mathcal{B}_{\text{L}}^{(3)} \circ \mathcal{B}_{\text{L}}^{(2)} \circ \mathcal{B}_{\text{L}}^{(1)} \circ E_{\text{mm}} \tag{2}$$

Grouping the repetitions with composition powers ($\mathcal{B}^{\circ n} = n$ consecutive blocks of the same type, each with its own weights) gives the compact form

$$f_\theta = \Psi \circ \Big(\big(\mathcal{B}_{\text{F}} \circ \mathcal{B}_{\text{L}}^{\circ 3} \big)^{\circ 8} \Big) \circ E_{\text{mm}} \tag{3}$$

Definition 2.3 (Vision encoder)**.** The vision encoder is itself a composition of 27 blocks acting on a sequence of image patches of width 1152:

$$\text{Enc}_{\text{V}} = \text{Post} \circ \mathcal{B}_{\text{V}}^{\circ 27} \circ \text{PatchEmbed} \tag{4}$$

PatchEmbed turns the raw input into vectors and Post is the final norm / pooling / merging (Table 3).

Layer pattern.

Symbol	Block type	Count	Params / block	Checkpoint indices
$\mathcal{B}_{\text{L}}$	Linear-attention (Gated DeltaNet) block	24	218,407,104	0–2, 4–6, 8–10, 12–14, 16–18, 20–22, 24–26, 28–30
$\mathcal{B}_{\text{F}}$	Full (global) attention block	8	209,723,904	3, 7, …, 31 (every 4th)
$\mathcal{B}_{\text{V}}$	Vision encoder block	27	15,239,504	0–26

Language-model decoder (32 layers). layers 0–31: [$\mathcal{B}_{\text{L}}\times 3$, $\mathcal{B}_{\text{F}}$] repeated 8× (checkpoint indices, i.e. $\ell - 1$).

Vision encoder (27 layers). layers 0–26: [$\mathcal{B}_{\text{V}}$] repeated 27× (checkpoint indices, i.e. $\ell - 1$).

2.2 The layers, top to bottom

Every function in the composition above is defined here. Only the attention (or linear-attention) sub-layer mixes information *between* positions t. Every other sub-layer (norms, feed-forward networks, projections) acts on each token vector $h_t \in \mathbb{R}^d$ independently, e.g. $\text{FFN}(h)_t = \text{FFN}(h_t)$. All weights follow the convention $W \in \mathbb{R}^{d_{\text{out}}\times d_{\text{in}}}$, so a projection of a token vector is Wu (as in an affine layer $f_i(u) = A^{(i)}u + b^{(i)}$).

Definition 2.4 (Token embedding)**.** $E : \{1, \ldots, V\}^T \to \mathbb{R}^{T\times d}$ looks up one row of the embedding matrix $W_E \in \mathbb{R}^{V\times d}$ per token:

$$E(x)_t = W_E[x_t, :], \qquad t = 1, \ldots, T. \tag{5}$$

With images, the multimodal embedding replaces the placeholder-token rows of $E(x)$ by projected encoder outputs:

$$E_{\text{mm}}(x, \cdot) = \text{Merge}\Big(E(x),\ \text{Proj}_{\text{img}}\big(\,\text{Enc}_{\text{V}}(\text{img})\big)\Big). \tag{6}$$

Definition 2.5 (Normalisation)**.** $\text{RMSNorm}(u) = \gamma \odot u / \sqrt{\frac{1}{d}\sum_i u_i^2 + \epsilon}$, with a learned scale $\gamma \in \mathbb{R}^d$, applied to each token vector separately. It fixes the scale of a vector without changing its direction. Every norm in the model has its own γ, distinguished by a subscript (e.g. RMSNorm_1, $\text{RMSNorm}_{\text{final}}$). LN (LayerNorm, used in some encoder blocks) additionally subtracts the mean and adds a bias.

Definition 2.6 (Block $\mathcal{B}_{\text{L}}$: Linear-attention (Gated DeltaNet) block)**.** A decoder block of this type maps $h \in \mathbb{R}^{T\times 4096}$ to the same space. It is a composition of 2 residual sub-layers, each of the form $(\text{id} + g)(h) = h + g(h)$:

$$\mathcal{B}_{\text{L}} = (\text{id} + \text{FFN} \circ \text{RMSNorm}_2) \circ (\text{id} + \text{Mixer} \circ \text{RMSNorm}_1) \tag{7}$$

Step by step, the sub-layers are:

$$a = h + \text{Mixer}(\text{RMSNorm}_1(h)) \qquad \text{pre-norm + residual}$$

$$
\begin{aligned}
\mathcal{B}(h) &= a + \mathrm{FFN}(\mathrm{RMSNorm}_2(a)) && \text{pre-norm + residual}\\
[q,\, k,\, v] &= \mathrm{SiLU}\big(\,\mathrm{CausalConv1d}(W_{qkv}\,u)\big) && \text{short causal conv (kernel 4) over time}\\
z &= W_z\,u && \text{output gate}\\
\beta_t &= \sigma(W_b\,u_t) && \text{write strength in (0,1)}\\
g_t &= -e^{A_{\log}} \cdot \mathrm{softplus}(W_a\,u_t + b_{dt}) && \text{decay } (\le 0)\text{: how fast memory fades}\\
q_t,\, k_t &= \mathrm{L2Norm}(q_t),\ \mathrm{L2Norm}(k_t)\\
S_t &= e^{g_t}\,S_{t-1}\big(I - \beta_t k_t k_t^\top\big) + \beta_t\,v_t k_t^\top && \text{fixed-size memory per head (delta rule)}\\
o_t &= S_t\,q_t && \text{read memory with the query}\\
\mathrm{Mixer}(u) &= W_{\mathrm{out}}\big(\,\mathrm{RMSNorm}(o)\odot\mathrm{SiLU}(z)\big)\\
\mathrm{FFN}(u) &= W_{\mathrm{down}}\big(\,\mathrm{SiLU}(W_{\mathrm{gate}}\,u)\odot W_{\mathrm{up}}\,u\big) && \text{gated MLP, width 12288}
\end{aligned}
$$

Parameters of one block ($|\theta_\ell| = 218{,}407{,}104$): $K_{\mathrm{conv}} \in \mathbb{R}^{8192\times 1\times 4}$, $W_{\mathrm{out}} \in \mathbb{R}^{4096\times 4096}$, $W_{qkv} \in \mathbb{R}^{8192\times 4096}$, $W_z \in \mathbb{R}^{4096\times 4096}$, $W_b \in \mathbb{R}^{32\times 4096}$, $W_a \in \mathbb{R}^{32\times 4096}$, $W_{\mathrm{gate}} \in \mathbb{R}^{12288\times 4096}$, $W_{\mathrm{up}} \in \mathbb{R}^{12288\times 4096}$, $W_{\mathrm{down}} \in \mathbb{R}^{4096\times 12288}$, plus 8,384 parameters in norm scales, biases and other vectors. This type is used 24 times.

Definition 2.7 (Block $\mathcal{B}_{\mathrm{F}}$: Full (global) attention block)**.** A decoder block of this type maps $h \in \mathbb{R}^{T\times 4096}$ to the same space. It is a composition of 2 residual sub-layers, each of the form $(\mathrm{id} + g)(h) = h + g(h)$:

$$
\mathcal{B}_{\mathrm{F}} = (\mathrm{id} + \mathrm{FFN}\circ\mathrm{RMSNorm}_2)\circ(\mathrm{id} + \mathrm{Attn}\circ\mathrm{RMSNorm}_1) \tag{8}
$$

Step by step, the sub-layers are:

$$
\begin{aligned}
a &= h + \mathrm{Attn}(\mathrm{RMSNorm}_1(h)) && \text{pre-norm + residual}\\
\mathcal{B}(h) &= a + \mathrm{FFN}(\mathrm{RMSNorm}_2(a)) && \text{pre-norm + residual}\\
[Q,\, G] &= W_Q\,u && \text{query and an output gate}\\
K,\ V &= W_K\,u,\quad W_V\,u && H_{kv} = 4 \text{ key/value heads}\\
Q,\ K &= \mathrm{RMSNorm}_q(Q),\ \mathrm{RMSNorm}_k(K) && \text{per-head normalisation}\\
Q,\ K &= \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K) && \text{rotate by position}\\
S_h &= s\,Q_h K_{g(h)}^\top + M && \text{mask } M\text{: causal}\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\,V_{g(h)}\\
\mathrm{Attn}(u) &= W_O\big([O_1;\ldots;O_H]\odot\sigma(G)\big)\\
\mathrm{FFN}(u) &= W_{\mathrm{down}}\big(\,\mathrm{SiLU}(W_{\mathrm{gate}}\,u)\odot W_{\mathrm{up}}\,u\big) && \text{gated MLP, width 12288}
\end{aligned}
$$

Parameters of one block ($|\theta_\ell| = 209{,}723{,}904$): $W_Q \in \mathbb{R}^{8192\times 4096}$, $W_K \in \mathbb{R}^{1024\times 4096}$, $W_V \in \mathbb{R}^{1024\times 4096}$, $W_O \in \mathbb{R}^{4096\times 4096}$, $W_{\mathrm{gate}} \in \mathbb{R}^{12288\times 4096}$, $W_{\mathrm{up}} \in \mathbb{R}^{12288\times 4096}$, $W_{\mathrm{down}} \in \mathbb{R}^{4096\times 12288}$, plus 8,704 parameters in norm scales, biases and other vectors. This type is used 8 times.

Definition 2.8 (Block $\mathcal{B}_{\mathrm{V}}$: Vision encoder block)**.** A vision-encoder block of this type maps $h \in \mathbb{R}^{T\times 1152}$ to the same space. It is a composition of 2 residual sub-layers, each of the form $(\mathrm{id} + g)(h) = h + g(h)$:

$$
\mathcal{B}_{\mathrm{V}} = (\mathrm{id} + \mathrm{FFN}\circ\mathrm{LN}_2)\circ(\mathrm{id} + \mathrm{Attn}\circ\mathrm{LN}_1) \tag{9}
$$

Step by step, the sub-layers are:

$$
\begin{aligned}
a &= h + \mathrm{Attn}(\mathrm{LN}_1(h)) && \text{pre-norm + residual}\\
\mathcal{B}(h) &= a + \mathrm{FFN}(\mathrm{LN}_2(a)) && \text{pre-norm + residual}\\
[Q,\, K,\, V] &= W_{qkv}\,u + b_{qkv} && \text{one fused projection}\\
Q,\ K &= \mathrm{RoPE}_{2D}(Q),\ \mathrm{RoPE}_{2D}(K) && \text{rotate by (row, column) patch position}\\
S_h &= s\,Q_h K_h^\top + M && \text{mask } M\text{: bidirectional}\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\,V_h\\
\mathrm{Attn}(u) &= W_O\,[O_1;\ldots;O_H] + b_O\\
\mathrm{FFN}(u) &= W_2\ \mathrm{GELU}_{\tanh}(W_1 u + b_1) + b_2 && \text{MLP, width 4304}
\end{aligned}
$$

Parameters of one block ($|\theta_\ell| = 15{,}239{,}504$): $W_{qkv} \in \mathbb{R}^{3456\times 1152}$, $W_O \in \mathbb{R}^{1152\times 1152}$, $W_1 \in \mathbb{R}^{4304\times 1152}$, $W_2 \in \mathbb{R}^{1152\times 4304}$, plus 14,672 parameters in norm scales, biases and other vectors. This type is used 27 times.

Definition 2.9 (Output head)**.** $\Psi : \mathbb{R}^{T\times d} \to \mathbb{R}^{T\times V}$ normalises the last hidden state and scores every vocabulary token with its own matrix $W_{\mathrm{LM}} \in \mathbb{R}^{V\times d}$:

$$
\Psi(h)_t = W_{\mathrm{LM}}\ \mathrm{RMSNorm}_{\mathrm{final}}(h)_t \tag{10}
$$

2.3 Neurons and attention heads

Definition 2.10 (Neuron, pre-activation value and state)**.** The feed-forward sub-layer of block ℓ contains $d_{ff}^{(\ell)} = 12288$ neurons. Neuron j of layer ℓ is the scalar function of a normalised token vector $u \in \mathbb{R}^d$

$$\eta_j^{(\ell)}(u) = \mathrm{SiLU}\left(W_{\mathrm{gate},j}^{(\ell)} \cdot u\right)\left(W_{\mathrm{up},j}^{(\ell)} \cdot u\right), \tag{11}$$

where $W_{\mathrm{gate},j}$ is the j-th row of W_{gate}. The feed-forward output is the sum of the neuron outputs, each written along its own column of W_{down}:

$$\mathrm{FFN}_\ell(u) = \sum_{j=1}^{d_{ff}^{(\ell)}} \eta_j^{(\ell)}(u)\, W_{\mathrm{down}}^{(\ell)}[:, j]. \tag{12}$$

Across its L blocks the network has $N = \sum_{\ell=1}^{L} d_{ff}^{(\ell)} = 393{,}216$ feed-forward neurons. For input x at position t, with $u_t^{(\ell)}(x)$ the normalised input of the feed-forward sub-layer:

- $\mathcal{V}(\eta; x, t) = W_{\mathrm{gate},j}^{(\ell)} \cdot u_t^{(\ell)}(x) \in \mathbb{R}$ is the **pre-activation value** of neuron η (the argument of the non-linearity).
- $\mathcal{S}(\eta; x, t)$ is its **activation state**: *active* if $\mathcal{V} > 0$ and *inactive* if $\mathcal{V} \le 0$. The non-linearity is smooth, unlike ReLU, so an inactive neuron's output is strongly suppressed rather than exactly zero.

Definition 2.11 (Attention head and attention pattern)**.** Head h of an attention layer ℓ reads from the whole sequence through its attention pattern $P_h^{(\ell)}(x) \in \mathbb{R}^{T \times T}$ (row t = how much position t attends to each earlier position) and writes through its own $d \times d_h$ slice $W_{O,h}^{(\ell)}$ of W_O:

$$\mathrm{Attn}_\ell(u) = \sum_{h=1}^{H} \left(P_h^{(\ell)} V_{g(h)} \odot \sigma(G_h)\right) W_{O,h}^{(\ell)\top}. \tag{13}$$

$g(h)$ is the key/value head shared by query head h (grouped-query attention). The pattern P_h plays the role that the activation state plays for neurons: once P_h is fixed, the head is linear in its values. The network has 8 attention layers $\times$ $H = 16$ heads $= 128$ attention heads.

2.4 Architecture and parameters

Definition 2.12 (Architecture and parameters)**.** The **architecture** is the tuple of sizes and the layer-type sequence:

$$\mathcal{A} = (V, d, L, c, H, H_{kv}, d_h, d_{ff}) = (248{,}320,\ 4096,\ 32,\ ((\mathrm{L}^3\,\mathrm{F})^8),\ 16,\ 4,\ 256,\ 12288). \tag{14}$$

The **parameters** are $\theta = \{W_E,\ \theta_1, \ldots, \theta_L,\ \gamma_{\mathrm{final}},\ W_{\mathrm{LM}},\ \theta_{\mathrm{Enc_V}},\ \theta_{\mathrm{Proj}}\}$. Grouping the blocks by type,

$$\begin{aligned} |\theta| &= \sum_{\text{types } c} n_c\, |\theta_c| + |\theta_{\mathrm{rest}}| \\ &= 24 \times 218{,}407{,}104 + 8 \times 209{,}723{,}904 + 27 \times 15{,}239{,}504 \\ &\quad + 2{,}078{,}785{,}408 \\ &= 9{,}409{,}813{,}744, \end{aligned}$$

where n_c is how often type c occurs and θ_{rest} holds everything outside the repeated stacks (embedding, final norm, encoder stems and projector, ...; Table 3).

2.5 Output representations

Definition 2.13 (Output representations)**.** For an input sequence x, the output can be read at three levels:

1. **Logits:** $y = f_\theta(x) \in \mathbb{R}^{T \times V}$, one score per vocabulary token at every position.
2. **Next-token distribution:** $p(x)_{t,k} = \dfrac{\exp(y_{t,k})}{\sum_{j=1}^{V} \exp(y_{t,j})} = \Pr\left(x_{t+1} = k \mid x_1, \ldots, x_t\right)$.
3. **Predicted next token:** $z(x) = \mathrm{argmax}_{k \in \{1,\ldots,V\}}\, y_{T,k}$ (greedy decoding). Generation appends $z(x)$ (or a sample from $p(x)_T$) to x and applies f_θ again; in practice the keys and values of earlier positions are cached.

The model is **causal**: y_t depends only on $x_1, \ldots, x_t$, because every attention mask hides later positions. Training minimises the cross-entropy $-\sum_t \log p(x)_{t, x_{t+1}}$.

2.6 Intermediate states and sub-network decomposition

To study what happens at one layer $\ell \in \{1, \dots, L\}$, split the composition into a prefix, the layer itself and a suffix:

$$f_\theta = \underbrace{\Psi \circ \mathcal{B}^{(L)} \circ \cdots \circ \mathcal{B}^{(\ell+1)}}_{G_{\ell+1}\ \text{(suffix)}} \circ \underbrace{\mathcal{B}^{(\ell)}}_{\text{layer } \ell} \circ \underbrace{\mathcal{B}^{(\ell-1)} \circ \cdots \circ \mathcal{B}^{(1)} \circ E_{\text{mm}}}_{F_{\ell-1}\ \text{(prefix)}} \tag{15}$$

- **Prefix** $F_{\ell-1} : \mathcal{X} \to \mathbb{R}^{T \times d}$ maps the tokens to the hidden state entering layer ℓ ($F_0 = E_{\text{mm}}$).
- **Hidden state** (residual stream) after layer ℓ: $h^{(\ell)} = F_\ell(x) = \mathcal{B}^{(\ell)}(h^{(\ell-1)}) \in \mathbb{R}^{T \times d}$, with $h^{(0)} = E_{\text{mm}}(x)$.
- **Inside layer ℓ:** the attention write $a^{(\ell)} = \text{Mixer}_\ell\big(\text{RMSNorm}_1(h^{(\ell-1)})\big)$, the mid-block state $\hat{h}^{(\ell)} = h^{(\ell-1)} + a^{(\ell)}$, the feed-forward input $u^{(\ell)} = \text{RMSNorm}_2(\hat{h}^{(\ell)})$, the pre-activations $\hat{x}^{(\ell)} = u^{(\ell)} W_{\text{gate}}^{(\ell)\top} \in \mathbb{R}^{T \times d_{ff}}$ and the neuron outputs $\eta^{(\ell)}(u^{(\ell)}) \in \mathbb{R}^{T \times d_{ff}}$, and the feed-forward write $m^{(\ell)} = \text{FFN}_\ell(u^{(\ell)})$.
- **Suffix** $G_{\ell+1} : \mathbb{R}^{T \times d} \to \mathbb{R}^{T \times V}$ maps $h^{(\ell)}$ to the logits ($G_{L+1} = \Psi$).

Residual stream. Because every sub-layer only *adds* to the hidden state, unrolling the composition gives the final hidden state as the embedding plus everything the layers wrote:

$$h^{(L)} = h^{(0)} + \sum_{\ell=1}^{L} \big(a^{(\ell)} + m^{(\ell)}\big) \tag{16}$$

So $f_\theta(x) = \Psi(h^{(L)})$: each layer reads the current stream and adds a correction to it.

2.7 Local linear representation

A ReLU network is piecewise affine: inside a polytope cell it collapses to one matrix. This model is not, because softmax, SiLU and the normalisations are smooth. Its local behaviour is described by the first-order (Jacobian) linearisation instead. Tokens are discrete, so we perturb the embedded input $h^{(0)}$.

Definition 2.14 (Local linear response)**.** Let $J_\ell = D\mathcal{B}^{(\ell)}(h^{(\ell-1)})$ be the Jacobian of block ℓ at the current hidden state (tensors flattened to vectors). The chain rule applied to the residual sub-layers $(\text{id} + g_{\ell,i})$ gives

$$J_\ell = (I + Dg_{\ell,2})(I + Dg_{\ell,1}), \tag{17}$$

where $g_{\ell,1}$ is the attention branch, $g_{\ell,2}$ the feed-forward branch and each Dg is taken at that branch's input. For a small perturbation δ of $h^{(0)}$, the response of the network decomposed around layer ℓ is

$$f_\theta(h^{(0)} + \delta) - f_\theta(h^{(0)}) \approx G_x^{(\ell+1)} \cdot J_\ell \cdot F_x^{(\ell-1)} \cdot \delta, \tag{18}$$

with $F_x^{(\ell-1)} = J_{\ell-1} \cdots J_1$ (prefix) and $G_x^{(\ell+1)} = J_\Psi\, J_L \cdots J_{\ell+1}$ (suffix). Multiplying out the products of $(I + Dg)$ terms writes $J_\Psi J_L \cdots J_1$ as a sum over *paths*: the identity path straight from embedding to head, plus one term for every subset of sub-layers a signal passes through.

2.8 Summary of notation

Symbol	Meaning
$f_\theta : \mathcal{X} \to \mathcal{Y}$	the whole model, $\{1..V\}^T \to \mathbb{R}^{T \times V}$
$E_{\text{mm}},\ \Psi$	embedding and output head
$\text{Enc}_\text{V},\ \text{Enc}_\text{A},\ \text{Proj}$	vision / audio encoders and their projectors into the text width
$\mathcal{B}^{(\ell)},\ \theta_\ell$	block ℓ and its parameters, $\ell = 1, \dots, L$
$\mathcal{B}_c,\ c(\ell)$	block type c; type of layer ℓ
$\mathcal{B}^{\circ n}$	n consecutive blocks of one type (each with its own weights)
$(\text{id} + g)$	residual sub-layer $h \mapsto h + g(h)$
$h^{(\ell)}$	hidden state (residual stream) after layer ℓ; $h^{(0)}$ = embeddings
$a^{(\ell)},\ m^{(\ell)}$	what the attention / feed-forward sub-layer of layer ℓ adds to the stream
$\eta_j^{(\ell)}$	j-th feed-forward neuron of layer ℓ
$\mathcal{V}(\eta; x, t),\ \mathcal{S}(\eta; x, t)$	pre-activation value and active/inactive state of a neuron

continued on next page

Symbol	Meaning
$P_h^{(\ell)}$	attention pattern of head h in layer ℓ ($T \times T$)
$F_{\ell-1},\ G_{\ell+1}$	prefix (embedding to layer $\ell-1$) and suffix (layer $\ell+1$ to logits)
$J_\ell,\ F_x^{(\ell-1)},\ G_x^{(\ell+1)}$	Jacobian of block ℓ, and of the prefix / suffix at input x
$p(x),\ z(x)$	next-token distribution and greedy next-token prediction

3 Forward Pass in Pseudocode

Algorithm 1 Qwen3.5-9B: full forward pass

Require: token ids x of shape $B \times T$; optional images / audio

$h \leftarrow E(x)$ ▷ $h^{(0)}$: one vector per token, $B{\times}T{\times}4096$
if image input present **then**
  $e \leftarrow \mathrm{Enc_V}(\text{image})$ ▷ vision encoder: 27 layers, Algorithm 2
  $h[\text{image token slots}] \leftarrow \mathrm{Proj}(e)$ ▷ project into text width 4096
end if
$\ell \leftarrow 0$
for $r \leftarrow 1$ **to** 8 **do** ▷ layers 1–32: pattern of 4 blocks, repeated 8 times
  for $j \leftarrow 1$ **to** 3 **do**
    $\ell \leftarrow \ell + 1;\ h \leftarrow \mathcal{B}_{\mathrm{L}}(h;\, \theta_\ell)$ ▷ Linear-attention (Gated DeltaNet) block
  end for
  $\ell \leftarrow \ell + 1;\ h \leftarrow \mathcal{B}_{\mathrm{F}}(h;\, \theta_\ell)$ ▷ Full (global) attention block
end for
$h \leftarrow \mathrm{RMSNorm}_{\mathrm{final}}(h)$ ▷ $h = h^{(L)}$; this and the next step are Ψ
$y \leftarrow h\, W_{\mathrm{LM}}^\top$ ▷ logits $B{\times}T{\times}248,320$
return y ▷ $\mathrm{softmax}(y)$ = next-token probabilities

Algorithm 2 Vision encoder

$h \leftarrow \mathrm{PatchEmbed}(\text{pixels}) + \text{position embedding}$ ▷ one vector per image patch
for $\ell \leftarrow 1$ **to** 27 **do** ▷ 27 blocks of type $\mathcal{B}_{\mathrm{V}}$
  $h \leftarrow \mathcal{B}_{\mathrm{V}}(h;\, \theta_\ell)$ ▷ Vision encoder block; result is $h^{(\ell)}$
end for
$h \leftarrow \mathrm{Post}(h)$ ▷ final norm / pooling / merging, see Table 3
return h

Algorithm 3 Block $\mathcal{B}_{\mathrm{L}}(h)$ — Linear-attention (Gated DeltaNet) block

Require: h of shape $B \times T \times 4096$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: Gated DeltaNet (linear attention, fixed-size memory)
$q, k, v \leftarrow \mathrm{split}(\mathrm{SiLU}(\mathrm{CausalConv1d}(uW_{qkv}^\top)))$ ▷ 16 q/k heads of 128, 32 v heads of 128
$z \leftarrow uW_z^\top;\ \beta \leftarrow \sigma(uW_b^\top);\ g \leftarrow -e^{A_{\log}}\,\mathrm{softplus}(uW_a^\top + b_{dt})$ ▷ gate, write strength, decay
for $t \leftarrow 1$ **to** T **do** ▷ conceptually sequential; computed in chunks
  $S \leftarrow e^{g_t} S\,(I - \beta_t k_t k_t^\top) + \beta_t v_t k_t^\top$ ▷ forget a bit, overwrite what k_t points at
  $o_t \leftarrow S\, q_t$ ▷ read
end for
$m \leftarrow \mathrm{RMSNorm}(o) \odot \mathrm{SiLU}(z)\ W_{\mathrm{out}}^\top$ ▷ back to $B{\times}T{\times}4096$
$h \leftarrow r + m$ ▷ add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$m \leftarrow \big(\mathrm{SiLU}(uW_{\mathrm{gate}}^\top) \odot uW_{\mathrm{up}}^\top\big) W_{\mathrm{down}}^\top$ ▷ expand to 12288, gate, shrink back to 4096
$h \leftarrow r + m$ ▷ add residual
return h

Algorithm 4 Block $\mathcal{B}_{\mathrm{F}}(h)$ — Full (global) attention block

Require: h of shape $B \times T \times 4096$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^\top$ ▷ $B{\times}T{\times}16{\times}256$ (16 query heads); second half is a gate G
$K \leftarrow uW_K^\top;\ V \leftarrow uW_V^\top$ ▷ $B{\times}T{\times}4{\times}256$ (4 key/value heads, each shared by 4 query heads)
$Q \leftarrow \mathrm{RMSNorm}(Q);\ K \leftarrow \mathrm{RMSNorm}(K)$ ▷ normalise each head vector (stabilises training)
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs
$P \leftarrow \mathrm{softmax}(s\,QK^\top + M)$ ▷ mask: causal: a token sees only earlier tokens
$O \leftarrow PV$ ▷ $B{\times}T{\times}16{\times}256$
$O \leftarrow O \odot \sigma(G)$ ▷ output gate decides how much of each head to keep
$m \leftarrow O\,W_O^\top$ ▷ concatenate heads, project back to $B{\times}T{\times}4096$
$h \leftarrow r + m$ ▷ add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$m \leftarrow \big(\mathrm{SiLU}(uW_{\mathrm{gate}}^\top) \odot uW_{\mathrm{up}}^\top\big)W_{\mathrm{down}}^\top$ ▷ expand to 12288, gate, shrink back to 4096
$h \leftarrow r + m$ ▷ add residual
return h

Algorithm 5 Block $\mathcal{B}_{\mathrm{V}}(h)$ — Vision encoder block

Require: h of shape $B \times T \times 1152$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{LN}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q, K, V \leftarrow \mathrm{split}(uW_{qkv}^\top + b)$ ▷ each $B{\times}T{\times}16{\times}72$
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ 2-D rotary position (row, column)
$P \leftarrow \mathrm{softmax}(s\,QK^\top + M)$ ▷ mask: every patch/frame sees every other
$O \leftarrow PV$ ▷ $B{\times}T{\times}16{\times}72$
$m \leftarrow O\,W_O^\top + b_O$ ▷ concatenate heads, project back to $B{\times}T{\times}1152$
$h \leftarrow r + m$ ▷ add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{LN}(h)$
$m \leftarrow \mathrm{GELU}_{\tanh}(uW_1^\top + b_1)W_2^\top + b_2$ ▷ expand to 4304, shrink back to 1152
$h \leftarrow r + m$ ▷ add residual
return h

4 Compact Architecture (repeated layers written once)

4.1 Modules outside the repeated stacks

Table 3: Weights that appear once

Weight	Type	Shape	Params
`model.visual.patch_embed.proj.weight`	Conv3d	$1152 \times 3 \times 2 \times 16 \times 16$	1,769,472
`model.visual.patch_embed.proj.bias`	Conv3d	1152	1,152
`model.visual.pos_embed.weight`	Embedding	2304×1152	2,654,208
`model.visual.merger.norm.weight`	LayerNorm	1152	1,152
`model.visual.merger.norm.bias`	LayerNorm	1152	1,152
`model.visual.merger.linear_fc1.weight`	Linear	4608×4608	21,233,664
`model.visual.merger.linear_fc1.bias`	Linear	4608	4,608
`model.visual.merger.linear_fc2.weight`	Linear	4096×4608	18,874,368
`model.visual.merger.linear_fc2.bias`	Linear	4096	4,096
`model.language_model.embed_tokens.weight`	Embedding	248320×4096	1,017,118,720
`model.language_model.norm.weight`	Qwen3_5RMSNorm	4096	4,096
`lm_head.weight`	Linear	248320×4096	1,017,118,720

4.2 Language-model decoder: `model.language_model.layers`

Table 4: $\mathcal{B}_{\mathrm{L}}$: Linear-attention (Gated DeltaNet) block — layers 0–2, 4–6, 8–10, 12–14, 16–18, 20–22, 24–26, 28–30; weight names are relative to `model.language_model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`linear_attn.dt_bias`	Qwen3_5GatedDeltaNet	32	32
`linear_attn.A_log`	Qwen3_5GatedDeltaNet	32	32
`linear_attn.conv1d.weight`	Conv1d	$8192 \times 1 \times 4$	32,768
`linear_attn.norm.weight`	Qwen3_5RMSNormGated	128	128
`linear_attn.out_proj.weight`	Linear	4096×4096	16,777,216
`linear_attn.in_proj_qkv.weight`	Linear	8192×4096	33,554,432
`linear_attn.in_proj_z.weight`	Linear	4096×4096	16,777,216
`linear_attn.in_proj_b.weight`	Linear	32×4096	131,072
`linear_attn.in_proj_a.weight`	Linear	32×4096	131,072
`mlp.gate_proj.weight`	Linear	12288×4096	50,331,648
`mlp.up_proj.weight`	Linear	12288×4096	50,331,648
`mlp.down_proj.weight`	Linear	4096×12288	50,331,648
`input_layernorm.weight`	Qwen3_5RMSNorm	4096	4,096
`post_attention_layernorm.weight`	Qwen3_5RMSNorm	4096	4,096
Per block			**218,407,104**
× 24 blocks			**5,241,770,496**

Table 5: $\mathcal{B}_{\mathrm{F}}$: Full (global) attention block — layers 3, 7, …, 31 (every 4th); weight names are relative to `model.language_model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.q_proj.weight`	Linear	8192×4096	33,554,432
`self_attn.k_proj.weight`	Linear	1024×4096	4,194,304
`self_attn.v_proj.weight`	Linear	1024×4096	4,194,304
`self_attn.o_proj.weight`	Linear	4096×4096	16,777,216
`self_attn.q_norm.weight`	Qwen3_5RMSNorm	256	256
`self_attn.k_norm.weight`	Qwen3_5RMSNorm	256	256
`mlp.gate_proj.weight`	Linear	12288×4096	50,331,648
`mlp.up_proj.weight`	Linear	12288×4096	50,331,648
`mlp.down_proj.weight`	Linear	4096×12288	50,331,648
`input_layernorm.weight`	Qwen3_5RMSNorm	4096	4,096
`post_attention_layernorm.weight`	Qwen3_5RMSNorm	4096	4,096
Per block			**209,723,904**
× 8 blocks			**1,677,791,232**

4.3 Vision encoder: `model.visual.blocks`

Table 6: $\mathcal{B}_{\mathrm{V}}$: Vision encoder block — layers 0–26; weight names are relative to `model.visual.blocks.`ℓ

Weight (relative)	Type	Shape	Params
`norm1.weight`	LayerNorm	1152	1,152
`norm1.bias`	LayerNorm	1152	1,152
`norm2.weight`	LayerNorm	1152	1,152
`norm2.bias`	LayerNorm	1152	1,152
`attn.qkv.weight`	Linear	3456×1152	3,981,312

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Weight (relative)	Type	Shape	Params
`attn.qkv.bias`	Linear	3456	3,456
`attn.proj.weight`	Linear	1152×1152	1,327,104
`attn.proj.bias`	Linear	1152	1,152
`mlp.linear_fc1.weight`	Linear	4304×1152	4,958,208
`mlp.linear_fc1.bias`	Linear	4304	4,304
`mlp.linear_fc2.weight`	Linear	1152×4304	4,958,208
`mlp.linear_fc2.bias`	Linear	1152	1,152
Per block			**15,239,504**
× 27 blocks			**411,466,608**

4.4 Checkpoint-only modules

Multi-token-prediction (MTP) head (`mtp.*`). An extra one-layer transformer that predicts the token after next. It concatenates RMSNorm(embedding of token $t{+}1$) with RMSNorm(final hidden state at t), projects $2d \to d$ (`fc`), runs one full-attention decoder layer, then a norm and the shared LM head. Serving engines (vLLM/SGLang) use it for speculative decoding. The transformers class does not instantiate it, so it is not counted in the total above.

Tensor	Shape	Dtype	Params
`mtp.fc.weight`	4096×8192	bf16	33,554,432
`mtp.layers.0.input_layernorm.weight`	4096	bf16	4,096
`mtp.layers.0.mlp.down_proj.weight`	4096×12288	bf16	50,331,648
`mtp.layers.0.mlp.gate_proj.weight`	12288×4096	bf16	50,331,648
`mtp.layers.0.mlp.up_proj.weight`	12288×4096	bf16	50,331,648
`mtp.layers.0.post_attention_layernorm.weight`	4096	bf16	4,096
`mtp.layers.0.self_attn.k_norm.weight`	256	bf16	256
`mtp.layers.0.self_attn.k_proj.weight`	1024×4096	bf16	4,194,304
`mtp.layers.0.self_attn.o_proj.weight`	4096×4096	bf16	16,777,216
`mtp.layers.0.self_attn.q_norm.weight`	256	bf16	256
`mtp.layers.0.self_attn.q_proj.weight`	8192×4096	bf16	33,554,432
`mtp.layers.0.self_attn.v_proj.weight`	1024×4096	bf16	4,194,304
`mtp.norm.weight`	4096	bf16	4,096
`mtp.pre_fc_norm_embedding.weight`	4096	bf16	4,096
`mtp.pre_fc_norm_hidden.weight`	4096	bf16	4,096

5 What Each Layer Does

One row per *kind* of layer present in this model. "Appears as" gives the module names (layer indices removed).

Layer (appears as)	What it does
LANGUAGE MODEL	
Token embedding `embed_tokens`	Lookup table with one learned vector per vocabulary token. Turns token ids into the vectors the network computes with (Gemma multiplies them by $\sqrt{d}$).
Query projection (q_proj) `self_attn.q_proj`	Turns each token vector into query vectors, one per head: "what am I looking for?". Here it also outputs an equally sized gate for the attention output (hence the doubled width).
Query norm (q_norm) `self_attn.q_norm`	RMSNorm applied to every query head vector before RoPE. Keeps attention logits in a sane range (QK-norm).
Key projection (k_proj) `self_attn.k_proj`	Turns each token into key vectors: "what do I contain?". Queries are compared with keys to decide where to attend. With grouped-query attention there are fewer key heads than query heads.
Key norm (k_norm) `self_attn.k_norm`	RMSNorm applied to every key head vector before RoPE.
Value projection (v_proj) `self_attn.v_proj`	Turns each token into value vectors: the information actually copied to tokens that attend to it.

continued on next page

Layer (appears as)	What it does
DeltaNet output projection `linear_attn.out_proj`	Mixes the heads back into one hidden-size vector.
Output projection (o_proj) `self_attn.o_proj`	Concatenates all heads' results and mixes them back into one hidden-size vector per token.
LM head (output projection) `lm_head`	Maps each final hidden vector to one score (logit) per vocabulary token. Softmax of the logits gives the next-token probabilities. When tied, it reuses the embedding matrix.
DeltaNet Q/K/V projection `linear_attn.in_proj_qkv`	Produces queries, keys and values for the linear-attention layer in one matrix.
DeltaNet output-gate projection `linear_attn.in_proj_z`	Produces gate z. The normalised output is multiplied by SiLU(z).
DeltaNet write-strength projection `linear_attn.in_proj_b`	Produces $\beta = \sigma(\cdot)$ per head: how strongly the current token overwrites the memory.
DeltaNet decay projection `linear_attn.in_proj_a`	Produces the input-dependent forgetting rate (with A_log and dt_bias).
Short causal convolution `linear_attn.conv1d`	Depthwise 1-D convolution over the last 4 tokens of Q/K/V. Gives cheap local context before the recurrent memory update.
A_log (decay base) `layers.linear_attn`	Per-head learned log decay rate. $-e^{A_{\log}}$ sets how quickly old memory fades.
dt_bias (decay bias) `layers.linear_attn`	Per-head bias added before the softplus that controls the decay step.
Gated output RMSNorm `linear_attn.norm`	Normalises each head's memory read-out and multiplies it by SiLU(z).
Gate projection (gate_proj) `mlp.gate_proj`	Expands the token vector to the MLP width and passes it through SiLU/GELU. The result acts as a soft on/off switch for each hidden unit.
Up projection (up_proj) `mlp.up_proj`	Expands the token vector to the MLP width (the "content" path). It is multiplied element-wise by the gate (SwiGLU/GeGLU).
Down projection (down_proj) `mlp.down_proj`	Shrinks the gated hidden vector back to the model width so it can be added to the residual stream.
Input norm (input_layernorm) `layers.input_layernorm`	RMSNorm applied to the residual stream before attention. Rescales each token vector to unit RMS, then multiplies by a learned per-channel weight.
post_attention_layernorm `layers.post_attention_layernorm`	Despite the name, this is the pre-MLP norm: RMSNorm on the residual stream after attention, feeding the MLP.
Final norm `norm`	RMSNorm applied once after the last decoder layer, just before the LM head.
VISION ENCODER	
3-D patch embedding (Conv3d) `patch_embed.proj`	Cuts images/videos into 2-frame × 16 × 16-pixel cubes and maps each cube to one vector.
Position embedding `visual.pos_embed`	Learned position vectors, interpolated to the image's patch grid.
Pre-attention LayerNorm `blocks.norm1`	LayerNorm before vision self-attention.
Pre-MLP LayerNorm `blocks.norm2`	LayerNorm before the vision MLP.
Fused Q/K/V projection `attn.qkv`	One matrix that produces queries, keys and values at once (split afterwards).
Attention output projection `attn.proj`	Mixes the concatenated head outputs back to the model width.
MLP up-projection (fc1) `mlp.linear_fc1`	Expands each vector to the wider MLP size before the GELU non-linearity.
MLP down-projection (fc2) `mlp.linear_fc2`	Shrinks back to the model width.

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Layer (appears as)	What it does
Merger LayerNorm `merger.norm`	Normalises patch features before 2×2 neighbouring patches are merged.
Merger MLP layer 1 `merger.linear_fc1`	Takes 4 neighbouring patches concatenated (2×2 merge, so 4× fewer image tokens) and mixes them.
Merger MLP layer 2 (to text width) `merger.linear_fc2`	Projects merged patches into the language model's hidden size.

6 Parameter Count

Table 9: Parameters by component

Component	Parameters	% of total
MLP / MoE	4,831,838,208	51.3%
Attention / token mixer	2,087,461,376	22.2%
Token embedding	1,017,118,720	10.8%
LM head	1,017,118,720	10.8%
Vision encoder	456,010,480	4.8%
Norms	266,240	0.0%
Total	**9,409,813,744**	**100.0%**

Table 10: Parameters by block type

Block type	Count	Params / block	Total
$\mathcal{B}_{\mathrm{L}}$ Linear-attention (Gated DeltaNet) block	24	218,407,104	5,241,770,496
$\mathcal{B}_{\mathrm{F}}$ Full (global) attention block	8	209,723,904	1,677,791,232
$\mathcal{B}_{\mathrm{V}}$ Vision encoder block	27	15,239,504	411,466,608
Modules outside stacks	1	–	2,078,785,408
Total			**9,409,813,744**

Appendix A Full Layer-by-Layer Table

Every weight tensor with every repetition written out (the original table).

Layer path	Type	Shape	Dtype	Params
`lm_head.weight`	Linear	248320 × 4096	bfloat16	1,017,118,720
`model`	Qwen3_5Model	–	–	–
`model.language_model`	Qwen3_5TextModel	–	–	–
`model.language_model.embed_tokens.weight`	Embedding	248320 × 4096	bfloat16	1,017,118,720
`model.language_model.layers.0`	Qwen3_5DecoderLayer	–	–	–
`model.language_model.layers.0.input_layernorm.weight`	Qwen3_5RMSNorm	4096	bfloat16	4,096
`model.language_model.layers.0.linear_attn.A_log`	Qwen3_5GatedDeltaNet	32	bfloat16	32
`model.language_model.layers.0.linear_attn`	Qwen3_5GatedDeltaNet	–	–	–
`model.language_model.layers.0.linear_attn.conv1d.weight`	Conv1d	8192 × 1 × 4	bfloat16	32,768
`model.language_model.layers.0.linear_attn.dt_bias`	Qwen3_5GatedDeltaNet	32	bfloat16	32
`model.language_model.layers.0.linear_attn.in_proj_a.weight`	Linear	32 × 4096	bfloat16	131,072
`model.language_model.layers.0.linear_attn.in_proj_b.weight`	Linear	32 × 4096	bfloat16	131,072
`model.language_model.layers.0.linear_attn.in_proj_qkv.weight`	Linear	8192 × 4096	bfloat16	33,554,432
`model.language_model.layers.0.linear_attn.in_proj_z.weight`	Linear	4096 × 4096	bfloat16	16,777,216
`model.language_model.layers.0.linear_attn.norm.weight`	Qwen3_5RMSNorm Gated	128	bfloat16	128

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.0.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.0.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.0.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.0.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.0.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.0.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.1	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.1.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.1.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.1.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.1.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.1.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.1.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.1.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.1.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.1.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.1.linear_attn.norm.weight	Qwen3_5RMSNorm Gated	128	bfloat16	128
model.language_model.layers.1.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.1.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.1.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.1.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.1.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.1.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.2	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.2.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.2.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.2.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.2.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.2.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.2.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.2.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.2.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.2.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.2.linear_attn.norm.weight	Qwen3_5RMSNorm Gated	128	bfloat16	128
model.language_model.layers.2.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.2.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.2.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.2.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.2.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.2.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.3	Qwen3_5DecoderLayer	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.3.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.3.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.3.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.3.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.3.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.3.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.3.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.3.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.3.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.3.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.3.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.3.self_attn.q_proj.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.3.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.4	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.4.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.4.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.4.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.4.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.4.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.4.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.4.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.4.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.4.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.4.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.4.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.4.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.4.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.4.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.4.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.4.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.5	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.5.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.5.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.5.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.5.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.5.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.5.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.5.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.5.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.5.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.5.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.5.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.5.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.5.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.5.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.5.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.5.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.6	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.6.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.6.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.6.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.6.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.6.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.6.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.6.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.6.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.6.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.6.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.6.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.6.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.6.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.6.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.6.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.6.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.7	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.7.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.7.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.7.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.7.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.7.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.7.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.7.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.7.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.7.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.7.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.7.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.7.self_attn.q_proj.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.7.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.8	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.8.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.8.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.8.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.8.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.8.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.8.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.8.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.8.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.8.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.8.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.8.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.8.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.8.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.8.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.8.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.8.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.9	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.9.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.9.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.9.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.9.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.9.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.9.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.9.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.9.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.9.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.9.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.9.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.9.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.9.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.9.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.9.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.9.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.10	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.10.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.10.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.10.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.10.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.10.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.10.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.10.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.10.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.10.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.10.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.10.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.10.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.10.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.10.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.10.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.10.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.11	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.11.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.11.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.11.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.11.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.11.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.11.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.11.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.11.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.11.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.11.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.11.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.11.self_attn.q_proj.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.11.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.12	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.12.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.12.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.12.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.12.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.12.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.12.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.12.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.12.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.12.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.12.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.12.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.12.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.12.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.12.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.12.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.12.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.13	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.13.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.13.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.13.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.13.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.13.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.13.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.13.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.13.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.13.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.13.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.13.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.13.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.13.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.13.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.13.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.13.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.14	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.14.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.14.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.14.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.14.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.14.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.14.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.14.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.14.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.14.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.14.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.14.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.14.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.14.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.14.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.14.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.14.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.15	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.15.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.15.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.15.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.15.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.15.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.15.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.15.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.15.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.15.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.15.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.15.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.15.self_attn.q_proj.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.15.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.16	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.16.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.16.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.16.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.16.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.16.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.16.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.16.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.16.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.16.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.16.linear_attn.norm.weight	Qwen3_5RMSNorm Gated	128	bfloat16	128
model.language_model.layers.16.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.16.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.16.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.16.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.16.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.16.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.17	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.17.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.17.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.17.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.17.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.17.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.17.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.17.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.17.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.17.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.17.linear_attn.norm.weight	Qwen3_5RMSNorm Gated	128	bfloat16	128
model.language_model.layers.17.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.17.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.17.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.17.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.17.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.17.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.18	Qwen3_5DecoderLayer	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.18.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.18.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.18.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.18.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.18.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.18.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.18.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.18.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.18.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.18.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.18.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.18.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.18.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.18.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.18.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.18.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.19	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.19.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.19.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.19.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.19.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.19.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.19.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.19.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.19.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.19.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.19.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.19.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.19.self_attn.q_proj.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.19.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.20	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.20.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.20.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.20.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.20.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.20.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.20.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.20.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.20.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.20.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.20.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.20.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.20.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.20.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.20.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.20.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.20.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.21	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.21.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.21.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.21.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.21.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.21.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.21.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.21.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.21.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.21.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.21.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.21.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.21.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.21.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.21.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.21.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.21.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.22	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.22.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.22.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.22.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.22.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.22.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.22.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.22.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.22.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.22.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.22.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.22.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.22.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.22.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.22.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.22.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.22.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.23	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.23.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.23.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.23.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.23.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.23.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.23.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.23.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.23.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.23.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.23.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.23.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.23.self_attn.q_proj.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.23.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.24	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.24.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.24.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.24.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.24.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.24.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.24.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.24.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.24.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.24.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.24.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.24.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.24.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.24.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.24.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.24.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.24.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.25	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.25.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.25.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.25.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.25.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.25.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.25.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.25.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.25.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.25.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.25.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.25.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.25.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.25.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.25.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.25.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.25.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.26	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.26.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.26.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.26.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.26.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.26.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.26.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.26.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.26.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.26.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.26.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.26.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.26.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.26.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.26.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.26.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.26.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.27	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.27.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.27.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.27.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.27.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.27.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.27.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.27.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.27.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.27.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.27.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.27.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.27.self_attn.q_proj.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.27.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.28	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.28.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.28.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.28.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.28.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.28.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.28.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.28.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.28.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.28.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.28.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.28.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.28.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.28.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.28.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.28.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.28.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.29	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.29.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.29.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.29.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.29.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.29.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.29.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.29.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.29.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.29.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.29.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.29.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.29.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.29.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.29.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.29.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.29.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.30	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.30.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.30.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.30.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.30.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.30.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.30.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.30.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.30.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.30.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.30.linear_attn.norm.weight	Qwen3_5RMSNorm Gated	128	bfloat16	128
model.language_model.layers.30.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.30.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.30.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.30.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.30.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.30.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.31	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.31.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.31.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.31.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.31.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.31.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.31.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.31.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.31.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.31.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.31.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.31.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.31.self_attn.q_proj.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.31.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.norm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.visual	Qwen3_5VisionModel	–	–	–
model.visual.blocks.0	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.0.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.0.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.0.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.0.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.0.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.0.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.0.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.0.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.0.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.0.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.0.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.0.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.0.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.0.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.1	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.1.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.1.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.1.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.1.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.1.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.1.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.1.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.1.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.1.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.1.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.1.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.1.norm1.weight	LayerNorm	1152	bfloat16	1,152

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Layer path	Type	Shape	Dtype	Params
model.visual.blocks.1.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.1.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.2	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.2.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.2.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.2.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.2.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.2.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.2.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.2.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.2.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.2.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.2.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.2.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.2.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.2.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.2.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.3	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.3.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.3.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.3.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.3.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.3.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.3.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.3.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.3.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.3.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.3.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.3.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.3.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.3.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.3.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.4	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.4.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.4.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.4.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.4.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.4.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.4.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.4.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.4.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.4.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.4.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.4.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.4.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.4.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.4.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.5	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.5.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.5.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.5.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.5.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.5.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.5.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.5.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.5.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.5.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.5.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.5.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.5.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.5.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.5.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.6	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.6.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.6.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.6.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.6.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.6.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.6.mlp	Qwen3_5VisionMLP	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.visual.blocks.6.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.6.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.6.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.6.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.6.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.6.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.6.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.6.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.7	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.7.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.7.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.7.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.7.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.7.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.7.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.7.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.7.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.7.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.7.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.7.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.7.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.7.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.7.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.8	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.8.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.8.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.8.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.8.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.8.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.8.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.8.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.8.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.8.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.8.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.8.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.8.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.8.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.8.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.9	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.9.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.9.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.9.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.9.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.9.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.9.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.9.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.9.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.9.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.9.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.9.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.9.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.9.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.9.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.10	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.10.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.10.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.10.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.10.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.10.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.10.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.10.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.10.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.10.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.10.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.10.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.10.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.10.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.10.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.11	Qwen3_5VisionBlock	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.visual.blocks.11.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.11.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.11.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.11.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.11.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.11.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.11.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.11.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.11.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.11.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.11.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.11.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.11.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.11.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.12	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.12.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.12.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.12.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.12.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.12.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.12.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.12.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.12.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.12.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.12.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.12.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.12.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.12.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.12.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.13	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.13.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.13.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.13.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.13.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.13.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.13.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.13.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.13.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.13.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.13.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.13.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.13.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.13.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.13.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.14	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.14.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.14.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.14.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.14.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.14.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.14.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.14.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.14.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.14.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.14.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.14.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.14.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.14.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.14.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.15	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.15.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.15.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.15.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.15.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.15.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.15.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.15.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.15.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.15.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152

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Layer path	Type	Shape	Dtype	Params
model.visual.blocks.15.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.15.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.15.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.15.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.15.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.16	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.16.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.16.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.16.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.16.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.16.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.16.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.16.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.16.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.16.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.16.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.16.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.16.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.16.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.16.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.17	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.17.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.17.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.17.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.17.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.17.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.17.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.17.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.17.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.17.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.17.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.17.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.17.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.17.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.17.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.18	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.18.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.18.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.18.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.18.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.18.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.18.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.18.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.18.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.18.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.18.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.18.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.18.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.18.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.18.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.19	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.19.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.19.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.19.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.19.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.19.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.19.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.19.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.19.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.19.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.19.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.19.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.19.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.19.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.19.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.20	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.20.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.20.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.20.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104

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Layer path	Type	Shape	Dtype	Params
model.visual.blocks.20.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.20.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.20.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.20.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.20.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.20.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.20.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.20.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.20.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.20.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.20.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.21	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.21.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.21.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.21.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.21.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.21.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.21.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.21.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.21.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.21.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.21.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.21.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.21.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.21.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.21.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.22	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.22.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.22.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.22.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.22.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.22.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.22.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.22.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.22.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.22.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.22.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.22.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.22.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.22.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.22.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.23	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.23.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.23.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.23.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.23.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.23.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.23.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.23.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.23.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.23.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.23.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.23.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.23.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.23.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.23.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.24	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.24.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.24.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.24.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.24.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.24.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.24.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.24.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.24.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.24.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.24.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.24.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.24.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.24.norm2.bias	LayerNorm	1152	bfloat16	1,152

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Layer path	Type	Shape	Dtype	Params
model.visual.blocks.24.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.25	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.25.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.25.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.25.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.25.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.25.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.25.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.25.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.25.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.25.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.25.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.25.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.25.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.25.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.25.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.26	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.26.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.26.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.26.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.26.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.26.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.26.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.26.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.26.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.26.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.26.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.26.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.26.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.26.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.26.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.merger	Qwen3_5VisionPatch Merger	–	–	–
model.visual.merger.linear_fc1.bias	Linear	4608	bfloat16	4,608
model.visual.merger.linear_fc1.weight	Linear	4608 × 4608	bfloat16	21,233,664
model.visual.merger.linear_fc2.bias	Linear	4096	bfloat16	4,096
model.visual.merger.linear_fc2.weight	Linear	4096 × 4608	bfloat16	18,874,368
model.visual.merger.norm.bias	LayerNorm	1152	bfloat16	1,152
model.visual.merger.norm.weight	LayerNorm	1152	bfloat16	1,152
model.visual.patch_embed	Qwen3_5VisionPatch Embed	–	–	–
model.visual.patch_embed.proj.bias	Conv3d	1152	bfloat16	1,152
model.visual.patch_embed.proj.weight	Conv3d	1152 × 3 × 2 × 16 × 16	bfloat16	1,769,472
model.visual.pos_embed.weight	Embedding	2304 × 1152	bfloat16	2,654,208

Appendix B Per-Layer Parameter Count

Module	Params	Running total
lm_head	1,017,118,720	1,017,118,720
model.language_model.embed_tokens	1,017,118,720	2,034,237,440
model.language_model.layers.0.input_layernorm	4,096	2,034,241,536
model.language_model.layers.0.linear_attn	64	2,034,241,600
model.language_model.layers.0.linear_attn.conv1d	32,768	2,034,274,368
model.language_model.layers.0.linear_attn.in_proj_a	131,072	2,034,405,440
model.language_model.layers.0.linear_attn.in_proj_b	131,072	2,034,536,512
model.language_model.layers.0.linear_attn. in_proj_qkv	33,554,432	2,068,090,944
model.language_model.layers.0.linear_attn.in_proj_z	16,777,216	2,084,868,160
model.language_model.layers.0.linear_attn.norm	128	2,084,868,288
model.language_model.layers.0.linear_attn.out_proj	16,777,216	2,101,645,504
model.language_model.layers.0.mlp.down_proj	50,331,648	2,151,977,152
model.language_model.layers.0.mlp.gate_proj	50,331,648	2,202,308,800
model.language_model.layers.0.mlp.up_proj	50,331,648	2,252,640,448

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Module	Params	Running total
model.language_model.layers.0. post_attention_layernorm	4,096	2,252,644,544
model.language_model.layers.1.input_layernorm	4,096	2,252,648,640
model.language_model.layers.1.linear_attn	64	2,252,648,704
model.language_model.layers.1.linear_attn.conv1d	32,768	2,252,681,472
model.language_model.layers.1.linear_attn.in_proj_a	131,072	2,252,812,544
model.language_model.layers.1.linear_attn.in_proj_b	131,072	2,252,943,616
model.language_model.layers.1.linear_attn. in_proj_qkv	33,554,432	2,286,498,048
model.language_model.layers.1.linear_attn.in_proj_z	16,777,216	2,303,275,264
model.language_model.layers.1.linear_attn.norm	128	2,303,275,392
model.language_model.layers.1.linear_attn.out_proj	16,777,216	2,320,052,608
model.language_model.layers.1.mlp.down_proj	50,331,648	2,370,384,256
model.language_model.layers.1.mlp.gate_proj	50,331,648	2,420,715,904
model.language_model.layers.1.mlp.up_proj	50,331,648	2,471,047,552
model.language_model.layers.1. post_attention_layernorm	4,096	2,471,051,648
model.language_model.layers.2.input_layernorm	4,096	2,471,055,744
model.language_model.layers.2.linear_attn	64	2,471,055,808
model.language_model.layers.2.linear_attn.conv1d	32,768	2,471,088,576
model.language_model.layers.2.linear_attn.in_proj_a	131,072	2,471,219,648
model.language_model.layers.2.linear_attn.in_proj_b	131,072	2,471,350,720
model.language_model.layers.2.linear_attn. in_proj_qkv	33,554,432	2,504,905,152
model.language_model.layers.2.linear_attn.in_proj_z	16,777,216	2,521,682,368
model.language_model.layers.2.linear_attn.norm	128	2,521,682,496
model.language_model.layers.2.linear_attn.out_proj	16,777,216	2,538,459,712
model.language_model.layers.2.mlp.down_proj	50,331,648	2,588,791,360
model.language_model.layers.2.mlp.gate_proj	50,331,648	2,639,123,008
model.language_model.layers.2.mlp.up_proj	50,331,648	2,689,454,656
model.language_model.layers.2. post_attention_layernorm	4,096	2,689,458,752
model.language_model.layers.3.input_layernorm	4,096	2,689,462,848
model.language_model.layers.3.mlp.down_proj	50,331,648	2,739,794,496
model.language_model.layers.3.mlp.gate_proj	50,331,648	2,790,126,144
model.language_model.layers.3.mlp.up_proj	50,331,648	2,840,457,792
model.language_model.layers.3. post_attention_layernorm	4,096	2,840,461,888
model.language_model.layers.3.self_attn.k_norm	256	2,840,462,144
model.language_model.layers.3.self_attn.k_proj	4,194,304	2,844,656,448
model.language_model.layers.3.self_attn.o_proj	16,777,216	2,861,433,664
model.language_model.layers.3.self_attn.q_norm	256	2,861,433,920
model.language_model.layers.3.self_attn.q_proj	33,554,432	2,894,988,352
model.language_model.layers.3.self_attn.v_proj	4,194,304	2,899,182,656
model.language_model.layers.4.input_layernorm	4,096	2,899,186,752
model.language_model.layers.4.linear_attn	64	2,899,186,816
model.language_model.layers.4.linear_attn.conv1d	32,768	2,899,219,584
model.language_model.layers.4.linear_attn.in_proj_a	131,072	2,899,350,656
model.language_model.layers.4.linear_attn.in_proj_b	131,072	2,899,481,728
model.language_model.layers.4.linear_attn. in_proj_qkv	33,554,432	2,933,036,160
model.language_model.layers.4.linear_attn.in_proj_z	16,777,216	2,949,813,376
model.language_model.layers.4.linear_attn.norm	128	2,949,813,504
model.language_model.layers.4.linear_attn.out_proj	16,777,216	2,966,590,720
model.language_model.layers.4.mlp.down_proj	50,331,648	3,016,922,368
model.language_model.layers.4.mlp.gate_proj	50,331,648	3,067,254,016
model.language_model.layers.4.mlp.up_proj	50,331,648	3,117,585,664

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Module	Params	Running total
model.language_model.layers.4. post_attention_layernorm	4,096	3,117,589,760
model.language_model.layers.5.input_layernorm	4,096	3,117,593,856
model.language_model.layers.5.linear_attn	64	3,117,593,920
model.language_model.layers.5.linear_attn.conv1d	32,768	3,117,626,688
model.language_model.layers.5.linear_attn.in_proj_a	131,072	3,117,757,760
model.language_model.layers.5.linear_attn.in_proj_b	131,072	3,117,888,832
model.language_model.layers.5.linear_attn. in_proj_qkv	33,554,432	3,151,443,264
model.language_model.layers.5.linear_attn.in_proj_z	16,777,216	3,168,220,480
model.language_model.layers.5.linear_attn.norm	128	3,168,220,608
model.language_model.layers.5.linear_attn.out_proj	16,777,216	3,184,997,824
model.language_model.layers.5.mlp.down_proj	50,331,648	3,235,329,472
model.language_model.layers.5.mlp.gate_proj	50,331,648	3,285,661,120
model.language_model.layers.5.mlp.up_proj	50,331,648	3,335,992,768
model.language_model.layers.5. post_attention_layernorm	4,096	3,335,996,864
model.language_model.layers.6.input_layernorm	4,096	3,336,000,960
model.language_model.layers.6.linear_attn	64	3,336,001,024
model.language_model.layers.6.linear_attn.conv1d	32,768	3,336,033,792
model.language_model.layers.6.linear_attn.in_proj_a	131,072	3,336,164,864
model.language_model.layers.6.linear_attn.in_proj_b	131,072	3,336,295,936
model.language_model.layers.6.linear_attn. in_proj_qkv	33,554,432	3,369,850,368
model.language_model.layers.6.linear_attn.in_proj_z	16,777,216	3,386,627,584
model.language_model.layers.6.linear_attn.norm	128	3,386,627,712
model.language_model.layers.6.linear_attn.out_proj	16,777,216	3,403,404,928
model.language_model.layers.6.mlp.down_proj	50,331,648	3,453,736,576
model.language_model.layers.6.mlp.gate_proj	50,331,648	3,504,068,224
model.language_model.layers.6.mlp.up_proj	50,331,648	3,554,399,872
model.language_model.layers.6. post_attention_layernorm	4,096	3,554,403,968
model.language_model.layers.7.input_layernorm	4,096	3,554,408,064
model.language_model.layers.7.mlp.down_proj	50,331,648	3,604,739,712
model.language_model.layers.7.mlp.gate_proj	50,331,648	3,655,071,360
model.language_model.layers.7.mlp.up_proj	50,331,648	3,705,403,008
model.language_model.layers.7. post_attention_layernorm	4,096	3,705,407,104
model.language_model.layers.7.self_attn.k_norm	256	3,705,407,360
model.language_model.layers.7.self_attn.k_proj	4,194,304	3,709,601,664
model.language_model.layers.7.self_attn.o_proj	16,777,216	3,726,378,880
model.language_model.layers.7.self_attn.q_norm	256	3,726,379,136
model.language_model.layers.7.self_attn.q_proj	33,554,432	3,759,933,568
model.language_model.layers.7.self_attn.v_proj	4,194,304	3,764,127,872
model.language_model.layers.8.input_layernorm	4,096	3,764,131,968
model.language_model.layers.8.linear_attn	64	3,764,132,032
model.language_model.layers.8.linear_attn.conv1d	32,768	3,764,164,800
model.language_model.layers.8.linear_attn.in_proj_a	131,072	3,764,295,872
model.language_model.layers.8.linear_attn.in_proj_b	131,072	3,764,426,944
model.language_model.layers.8.linear_attn. in_proj_qkv	33,554,432	3,797,981,376
model.language_model.layers.8.linear_attn.in_proj_z	16,777,216	3,814,758,592
model.language_model.layers.8.linear_attn.norm	128	3,814,758,720
model.language_model.layers.8.linear_attn.out_proj	16,777,216	3,831,535,936
model.language_model.layers.8.mlp.down_proj	50,331,648	3,881,867,584
model.language_model.layers.8.mlp.gate_proj	50,331,648	3,932,199,232
model.language_model.layers.8.mlp.up_proj	50,331,648	3,982,530,880

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Module	Params	Running total
model.language_model.layers.8.post_attention_layernorm	4,096	3,982,534,976
model.language_model.layers.9.input_layernorm	4,096	3,982,539,072
model.language_model.layers.9.linear_attn	64	3,982,539,136
model.language_model.layers.9.linear_attn.conv1d	32,768	3,982,571,904
model.language_model.layers.9.linear_attn.in_proj_a	131,072	3,982,702,976
model.language_model.layers.9.linear_attn.in_proj_b	131,072	3,982,834,048
model.language_model.layers.9.linear_attn.in_proj_qkv	33,554,432	4,016,388,480
model.language_model.layers.9.linear_attn.in_proj_z	16,777,216	4,033,165,696
model.language_model.layers.9.linear_attn.norm	128	4,033,165,824
model.language_model.layers.9.linear_attn.out_proj	16,777,216	4,049,943,040
model.language_model.layers.9.mlp.down_proj	50,331,648	4,100,274,688
model.language_model.layers.9.mlp.gate_proj	50,331,648	4,150,606,336
model.language_model.layers.9.mlp.up_proj	50,331,648	4,200,937,984
model.language_model.layers.9.post_attention_layernorm	4,096	4,200,942,080
model.language_model.layers.10.input_layernorm	4,096	4,200,946,176
model.language_model.layers.10.linear_attn	64	4,200,946,240
model.language_model.layers.10.linear_attn.conv1d	32,768	4,200,979,008
model.language_model.layers.10.linear_attn.in_proj_a	131,072	4,201,110,080
model.language_model.layers.10.linear_attn.in_proj_b	131,072	4,201,241,152
model.language_model.layers.10.linear_attn.in_proj_qkv	33,554,432	4,234,795,584
model.language_model.layers.10.linear_attn.in_proj_z	16,777,216	4,251,572,800
model.language_model.layers.10.linear_attn.norm	128	4,251,572,928
model.language_model.layers.10.linear_attn.out_proj	16,777,216	4,268,350,144
model.language_model.layers.10.mlp.down_proj	50,331,648	4,318,681,792
model.language_model.layers.10.mlp.gate_proj	50,331,648	4,369,013,440
model.language_model.layers.10.mlp.up_proj	50,331,648	4,419,345,088
model.language_model.layers.10.post_attention_layernorm	4,096	4,419,349,184
model.language_model.layers.11.input_layernorm	4,096	4,419,353,280
model.language_model.layers.11.mlp.down_proj	50,331,648	4,469,684,928
model.language_model.layers.11.mlp.gate_proj	50,331,648	4,520,016,576
model.language_model.layers.11.mlp.up_proj	50,331,648	4,570,348,224
model.language_model.layers.11.post_attention_layernorm	4,096	4,570,352,320
model.language_model.layers.11.self_attn.k_norm	256	4,570,352,576
model.language_model.layers.11.self_attn.k_proj	4,194,304	4,574,546,880
model.language_model.layers.11.self_attn.o_proj	16,777,216	4,591,324,096
model.language_model.layers.11.self_attn.q_norm	256	4,591,324,352
model.language_model.layers.11.self_attn.q_proj	33,554,432	4,624,878,784
model.language_model.layers.11.self_attn.v_proj	4,194,304	4,629,073,088
model.language_model.layers.12.input_layernorm	4,096	4,629,077,184
model.language_model.layers.12.linear_attn	64	4,629,077,248
model.language_model.layers.12.linear_attn.conv1d	32,768	4,629,110,016
model.language_model.layers.12.linear_attn.in_proj_a	131,072	4,629,241,088
model.language_model.layers.12.linear_attn.in_proj_b	131,072	4,629,372,160
model.language_model.layers.12.linear_attn.in_proj_qkv	33,554,432	4,662,926,592
model.language_model.layers.12.linear_attn.in_proj_z	16,777,216	4,679,703,808

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Module	Params	Running total
model.language_model.layers.12.linear_attn.norm	128	4,679,703,936
model.language_model.layers.12.linear_attn.out_proj	16,777,216	4,696,481,152
model.language_model.layers.12.mlp.down_proj	50,331,648	4,746,812,800
model.language_model.layers.12.mlp.gate_proj	50,331,648	4,797,144,448
model.language_model.layers.12.mlp.up_proj	50,331,648	4,847,476,096
model.language_model.layers.12. post_attention_layernorm	4,096	4,847,480,192
model.language_model.layers.13.input_layernorm	4,096	4,847,484,288
model.language_model.layers.13.linear_attn	64	4,847,484,352
model.language_model.layers.13.linear_attn.conv1d	32,768	4,847,517,120
model.language_model.layers.13.linear_attn. in_proj_a	131,072	4,847,648,192
model.language_model.layers.13.linear_attn. in_proj_b	131,072	4,847,779,264
model.language_model.layers.13.linear_attn. in_proj_qkv	33,554,432	4,881,333,696
model.language_model.layers.13.linear_attn. in_proj_z	16,777,216	4,898,110,912
model.language_model.layers.13.linear_attn.norm	128	4,898,111,040
model.language_model.layers.13.linear_attn.out_proj	16,777,216	4,914,888,256
model.language_model.layers.13.mlp.down_proj	50,331,648	4,965,219,904
model.language_model.layers.13.mlp.gate_proj	50,331,648	5,015,551,552
model.language_model.layers.13.mlp.up_proj	50,331,648	5,065,883,200
model.language_model.layers.13. post_attention_layernorm	4,096	5,065,887,296
model.language_model.layers.14.input_layernorm	4,096	5,065,891,392
model.language_model.layers.14.linear_attn	64	5,065,891,456
model.language_model.layers.14.linear_attn.conv1d	32,768	5,065,924,224
model.language_model.layers.14.linear_attn. in_proj_a	131,072	5,066,055,296
model.language_model.layers.14.linear_attn. in_proj_b	131,072	5,066,186,368
model.language_model.layers.14.linear_attn. in_proj_qkv	33,554,432	5,099,740,800
model.language_model.layers.14.linear_attn. in_proj_z	16,777,216	5,116,518,016
model.language_model.layers.14.linear_attn.norm	128	5,116,518,144
model.language_model.layers.14.linear_attn.out_proj	16,777,216	5,133,295,360
model.language_model.layers.14.mlp.down_proj	50,331,648	5,183,627,008
model.language_model.layers.14.mlp.gate_proj	50,331,648	5,233,958,656
model.language_model.layers.14.mlp.up_proj	50,331,648	5,284,290,304
model.language_model.layers.14. post_attention_layernorm	4,096	5,284,294,400
model.language_model.layers.15.input_layernorm	4,096	5,284,298,496
model.language_model.layers.15.mlp.down_proj	50,331,648	5,334,630,144
model.language_model.layers.15.mlp.gate_proj	50,331,648	5,384,961,792
model.language_model.layers.15.mlp.up_proj	50,331,648	5,435,293,440
model.language_model.layers.15. post_attention_layernorm	4,096	5,435,297,536
model.language_model.layers.15.self_attn.k_norm	256	5,435,297,792
model.language_model.layers.15.self_attn.k_proj	4,194,304	5,439,492,096
model.language_model.layers.15.self_attn.o_proj	16,777,216	5,456,269,312
model.language_model.layers.15.self_attn.q_norm	256	5,456,269,568
model.language_model.layers.15.self_attn.q_proj	33,554,432	5,489,824,000
model.language_model.layers.15.self_attn.v_proj	4,194,304	5,494,018,304
model.language_model.layers.16.input_layernorm	4,096	5,494,022,400
model.language_model.layers.16.linear_attn	64	5,494,022,464
model.language_model.layers.16.linear_attn.conv1d	32,768	5,494,055,232

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Module	Params	Running total
model.language_model.layers.16.linear_attn.in_proj_a	131,072	5,494,186,304
model.language_model.layers.16.linear_attn.in_proj_b	131,072	5,494,317,376
model.language_model.layers.16.linear_attn.in_proj_qkv	33,554,432	5,527,871,808
model.language_model.layers.16.linear_attn.in_proj_z	16,777,216	5,544,649,024
model.language_model.layers.16.linear_attn.norm	128	5,544,649,152
model.language_model.layers.16.linear_attn.out_proj	16,777,216	5,561,426,368
model.language_model.layers.16.mlp.down_proj	50,331,648	5,611,758,016
model.language_model.layers.16.mlp.gate_proj	50,331,648	5,662,089,664
model.language_model.layers.16.mlp.up_proj	50,331,648	5,712,421,312
model.language_model.layers.16.post_attention_layernorm	4,096	5,712,425,408
model.language_model.layers.17.input_layernorm	4,096	5,712,429,504
model.language_model.layers.17.linear_attn	64	5,712,429,568
model.language_model.layers.17.linear_attn.conv1d	32,768	5,712,462,336
model.language_model.layers.17.linear_attn.in_proj_a	131,072	5,712,593,408
model.language_model.layers.17.linear_attn.in_proj_b	131,072	5,712,724,480
model.language_model.layers.17.linear_attn.in_proj_qkv	33,554,432	5,746,278,912
model.language_model.layers.17.linear_attn.in_proj_z	16,777,216	5,763,056,128
model.language_model.layers.17.linear_attn.norm	128	5,763,056,256
model.language_model.layers.17.linear_attn.out_proj	16,777,216	5,779,833,472
model.language_model.layers.17.mlp.down_proj	50,331,648	5,830,165,120
model.language_model.layers.17.mlp.gate_proj	50,331,648	5,880,496,768
model.language_model.layers.17.mlp.up_proj	50,331,648	5,930,828,416
model.language_model.layers.17.post_attention_layernorm	4,096	5,930,832,512
model.language_model.layers.18.input_layernorm	4,096	5,930,836,608
model.language_model.layers.18.linear_attn	64	5,930,836,672
model.language_model.layers.18.linear_attn.conv1d	32,768	5,930,869,440
model.language_model.layers.18.linear_attn.in_proj_a	131,072	5,931,000,512
model.language_model.layers.18.linear_attn.in_proj_b	131,072	5,931,131,584
model.language_model.layers.18.linear_attn.in_proj_qkv	33,554,432	5,964,686,016
model.language_model.layers.18.linear_attn.in_proj_z	16,777,216	5,981,463,232
model.language_model.layers.18.linear_attn.norm	128	5,981,463,360
model.language_model.layers.18.linear_attn.out_proj	16,777,216	5,998,240,576
model.language_model.layers.18.mlp.down_proj	50,331,648	6,048,572,224
model.language_model.layers.18.mlp.gate_proj	50,331,648	6,098,903,872
model.language_model.layers.18.mlp.up_proj	50,331,648	6,149,235,520
model.language_model.layers.18.post_attention_layernorm	4,096	6,149,239,616
model.language_model.layers.19.input_layernorm	4,096	6,149,243,712
model.language_model.layers.19.mlp.down_proj	50,331,648	6,199,575,360
model.language_model.layers.19.mlp.gate_proj	50,331,648	6,249,907,008
model.language_model.layers.19.mlp.up_proj	50,331,648	6,300,238,656
model.language_model.layers.19.post_attention_layernorm	4,096	6,300,242,752
model.language_model.layers.19.self_attn.k_norm	256	6,300,243,008

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Module	Params	Running total
model.language_model.layers.19.self_attn.k_proj	4,194,304	6,304,437,312
model.language_model.layers.19.self_attn.o_proj	16,777,216	6,321,214,528
model.language_model.layers.19.self_attn.q_norm	256	6,321,214,784
model.language_model.layers.19.self_attn.q_proj	33,554,432	6,354,769,216
model.language_model.layers.19.self_attn.v_proj	4,194,304	6,358,963,520
model.language_model.layers.20.input_layernorm	4,096	6,358,967,616
model.language_model.layers.20.linear_attn	64	6,358,967,680
model.language_model.layers.20.linear_attn.conv1d	32,768	6,359,000,448
model.language_model.layers.20.linear_attn.in_proj_a	131,072	6,359,131,520
model.language_model.layers.20.linear_attn.in_proj_b	131,072	6,359,262,592
model.language_model.layers.20.linear_attn.in_proj_qkv	33,554,432	6,392,817,024
model.language_model.layers.20.linear_attn.in_proj_z	16,777,216	6,409,594,240
model.language_model.layers.20.linear_attn.norm	128	6,409,594,368
model.language_model.layers.20.linear_attn.out_proj	16,777,216	6,426,371,584
model.language_model.layers.20.mlp.down_proj	50,331,648	6,476,703,232
model.language_model.layers.20.mlp.gate_proj	50,331,648	6,527,034,880
model.language_model.layers.20.mlp.up_proj	50,331,648	6,577,366,528
model.language_model.layers.20.post_attention_layernorm	4,096	6,577,370,624
model.language_model.layers.21.input_layernorm	4,096	6,577,374,720
model.language_model.layers.21.linear_attn	64	6,577,374,784
model.language_model.layers.21.linear_attn.conv1d	32,768	6,577,407,552
model.language_model.layers.21.linear_attn.in_proj_a	131,072	6,577,538,624
model.language_model.layers.21.linear_attn.in_proj_b	131,072	6,577,669,696
model.language_model.layers.21.linear_attn.in_proj_qkv	33,554,432	6,611,224,128
model.language_model.layers.21.linear_attn.in_proj_z	16,777,216	6,628,001,344
model.language_model.layers.21.linear_attn.norm	128	6,628,001,472
model.language_model.layers.21.linear_attn.out_proj	16,777,216	6,644,778,688
model.language_model.layers.21.mlp.down_proj	50,331,648	6,695,110,336
model.language_model.layers.21.mlp.gate_proj	50,331,648	6,745,441,984
model.language_model.layers.21.mlp.up_proj	50,331,648	6,795,773,632
model.language_model.layers.21.post_attention_layernorm	4,096	6,795,777,728
model.language_model.layers.22.input_layernorm	4,096	6,795,781,824
model.language_model.layers.22.linear_attn	64	6,795,781,888
model.language_model.layers.22.linear_attn.conv1d	32,768	6,795,814,656
model.language_model.layers.22.linear_attn.in_proj_a	131,072	6,795,945,728
model.language_model.layers.22.linear_attn.in_proj_b	131,072	6,796,076,800
model.language_model.layers.22.linear_attn.in_proj_qkv	33,554,432	6,829,631,232
model.language_model.layers.22.linear_attn.in_proj_z	16,777,216	6,846,408,448
model.language_model.layers.22.linear_attn.norm	128	6,846,408,576
model.language_model.layers.22.linear_attn.out_proj	16,777,216	6,863,185,792
model.language_model.layers.22.mlp.down_proj	50,331,648	6,913,517,440
model.language_model.layers.22.mlp.gate_proj	50,331,648	6,963,849,088
model.language_model.layers.22.mlp.up_proj	50,331,648	7,014,180,736

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Module	Params	Running total
model.language_model.layers.22. post_attention_layernorm	4,096	7,014,184,832
model.language_model.layers.23.input_layernorm	4,096	7,014,188,928
model.language_model.layers.23.mlp.down_proj	50,331,648	7,064,520,576
model.language_model.layers.23.mlp.gate_proj	50,331,648	7,114,852,224
model.language_model.layers.23.mlp.up_proj	50,331,648	7,165,183,872
model.language_model.layers.23. post_attention_layernorm	4,096	7,165,187,968
model.language_model.layers.23.self_attn.k_norm	256	7,165,188,224
model.language_model.layers.23.self_attn.k_proj	4,194,304	7,169,382,528
model.language_model.layers.23.self_attn.o_proj	16,777,216	7,186,159,744
model.language_model.layers.23.self_attn.q_norm	256	7,186,160,000
model.language_model.layers.23.self_attn.q_proj	33,554,432	7,219,714,432
model.language_model.layers.23.self_attn.v_proj	4,194,304	7,223,908,736
model.language_model.layers.24.input_layernorm	4,096	7,223,912,832
model.language_model.layers.24.linear_attn	64	7,223,912,896
model.language_model.layers.24.linear_attn.conv1d	32,768	7,223,945,664
model.language_model.layers.24.linear_attn. in_proj_a	131,072	7,224,076,736
model.language_model.layers.24.linear_attn. in_proj_b	131,072	7,224,207,808
model.language_model.layers.24.linear_attn. in_proj_qkv	33,554,432	7,257,762,240
model.language_model.layers.24.linear_attn. in_proj_z	16,777,216	7,274,539,456
model.language_model.layers.24.linear_attn.norm	128	7,274,539,584
model.language_model.layers.24.linear_attn.out_proj	16,777,216	7,291,316,800
model.language_model.layers.24.mlp.down_proj	50,331,648	7,341,648,448
model.language_model.layers.24.mlp.gate_proj	50,331,648	7,391,980,096
model.language_model.layers.24.mlp.up_proj	50,331,648	7,442,311,744
model.language_model.layers.24. post_attention_layernorm	4,096	7,442,315,840
model.language_model.layers.25.input_layernorm	4,096	7,442,319,936
model.language_model.layers.25.linear_attn	64	7,442,320,000
model.language_model.layers.25.linear_attn.conv1d	32,768	7,442,352,768
model.language_model.layers.25.linear_attn. in_proj_a	131,072	7,442,483,840
model.language_model.layers.25.linear_attn. in_proj_b	131,072	7,442,614,912
model.language_model.layers.25.linear_attn. in_proj_qkv	33,554,432	7,476,169,344
model.language_model.layers.25.linear_attn. in_proj_z	16,777,216	7,492,946,560
model.language_model.layers.25.linear_attn.norm	128	7,492,946,688
model.language_model.layers.25.linear_attn.out_proj	16,777,216	7,509,723,904
model.language_model.layers.25.mlp.down_proj	50,331,648	7,560,055,552
model.language_model.layers.25.mlp.gate_proj	50,331,648	7,610,387,200
model.language_model.layers.25.mlp.up_proj	50,331,648	7,660,718,848
model.language_model.layers.25. post_attention_layernorm	4,096	7,660,722,944
model.language_model.layers.26.input_layernorm	4,096	7,660,727,040
model.language_model.layers.26.linear_attn	64	7,660,727,104
model.language_model.layers.26.linear_attn.conv1d	32,768	7,660,759,872
model.language_model.layers.26.linear_attn. in_proj_a	131,072	7,660,890,944
model.language_model.layers.26.linear_attn. in_proj_b	131,072	7,661,022,016

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Module	Params	Running total
model.language_model.layers.26.linear_attn.in_proj_qkv	33,554,432	7,694,576,448
model.language_model.layers.26.linear_attn.in_proj_z	16,777,216	7,711,353,664
model.language_model.layers.26.linear_attn.norm	128	7,711,353,792
model.language_model.layers.26.linear_attn.out_proj	16,777,216	7,728,131,008
model.language_model.layers.26.mlp.down_proj	50,331,648	7,778,462,656
model.language_model.layers.26.mlp.gate_proj	50,331,648	7,828,794,304
model.language_model.layers.26.mlp.up_proj	50,331,648	7,879,125,952
model.language_model.layers.26.post_attention_layernorm	4,096	7,879,130,048
model.language_model.layers.27.input_layernorm	4,096	7,879,134,144
model.language_model.layers.27.mlp.down_proj	50,331,648	7,929,465,792
model.language_model.layers.27.mlp.gate_proj	50,331,648	7,979,797,440
model.language_model.layers.27.mlp.up_proj	50,331,648	8,030,129,088
model.language_model.layers.27.post_attention_layernorm	4,096	8,030,133,184
model.language_model.layers.27.self_attn.k_norm	256	8,030,133,440
model.language_model.layers.27.self_attn.k_proj	4,194,304	8,034,327,744
model.language_model.layers.27.self_attn.o_proj	16,777,216	8,051,104,960
model.language_model.layers.27.self_attn.q_norm	256	8,051,105,216
model.language_model.layers.27.self_attn.q_proj	33,554,432	8,084,659,648
model.language_model.layers.27.self_attn.v_proj	4,194,304	8,088,853,952
model.language_model.layers.28.input_layernorm	4,096	8,088,858,048
model.language_model.layers.28.linear_attn	64	8,088,858,112
model.language_model.layers.28.linear_attn.conv1d	32,768	8,088,890,880
model.language_model.layers.28.linear_attn.in_proj_a	131,072	8,089,021,952
model.language_model.layers.28.linear_attn.in_proj_b	131,072	8,089,153,024
model.language_model.layers.28.linear_attn.in_proj_qkv	33,554,432	8,122,707,456
model.language_model.layers.28.linear_attn.in_proj_z	16,777,216	8,139,484,672
model.language_model.layers.28.linear_attn.norm	128	8,139,484,800
model.language_model.layers.28.linear_attn.out_proj	16,777,216	8,156,262,016
model.language_model.layers.28.mlp.down_proj	50,331,648	8,206,593,664
model.language_model.layers.28.mlp.gate_proj	50,331,648	8,256,925,312
model.language_model.layers.28.mlp.up_proj	50,331,648	8,307,256,960
model.language_model.layers.28.post_attention_layernorm	4,096	8,307,261,056
model.language_model.layers.29.input_layernorm	4,096	8,307,265,152
model.language_model.layers.29.linear_attn	64	8,307,265,216
model.language_model.layers.29.linear_attn.conv1d	32,768	8,307,297,984
model.language_model.layers.29.linear_attn.in_proj_a	131,072	8,307,429,056
model.language_model.layers.29.linear_attn.in_proj_b	131,072	8,307,560,128
model.language_model.layers.29.linear_attn.in_proj_qkv	33,554,432	8,341,114,560
model.language_model.layers.29.linear_attn.in_proj_z	16,777,216	8,357,891,776
model.language_model.layers.29.linear_attn.norm	128	8,357,891,904
model.language_model.layers.29.linear_attn.out_proj	16,777,216	8,374,669,120
model.language_model.layers.29.mlp.down_proj	50,331,648	8,425,000,768
model.language_model.layers.29.mlp.gate_proj	50,331,648	8,475,332,416
model.language_model.layers.29.mlp.up_proj	50,331,648	8,525,664,064

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Module	Params	Running total
model.language_model.layers.29. post_attention_layernorm	4,096	8,525,668,160
model.language_model.layers.30.input_layernorm	4,096	8,525,672,256
model.language_model.layers.30.linear_attn	64	8,525,672,320
model.language_model.layers.30.linear_attn.conv1d	32,768	8,525,705,088
model.language_model.layers.30.linear_attn. in_proj_a	131,072	8,525,836,160
model.language_model.layers.30.linear_attn. in_proj_b	131,072	8,525,967,232
model.language_model.layers.30.linear_attn. in_proj_qkv	33,554,432	8,559,521,664
model.language_model.layers.30.linear_attn. in_proj_z	16,777,216	8,576,298,880
model.language_model.layers.30.linear_attn.norm	128	8,576,299,008
model.language_model.layers.30.linear_attn.out_proj	16,777,216	8,593,076,224
model.language_model.layers.30.mlp.down_proj	50,331,648	8,643,407,872
model.language_model.layers.30.mlp.gate_proj	50,331,648	8,693,739,520
model.language_model.layers.30.mlp.up_proj	50,331,648	8,744,071,168
model.language_model.layers.30. post_attention_layernorm	4,096	8,744,075,264
model.language_model.layers.31.input_layernorm	4,096	8,744,079,360
model.language_model.layers.31.mlp.down_proj	50,331,648	8,794,411,008
model.language_model.layers.31.mlp.gate_proj	50,331,648	8,844,742,656
model.language_model.layers.31.mlp.up_proj	50,331,648	8,895,074,304
model.language_model.layers.31. post_attention_layernorm	4,096	8,895,078,400
model.language_model.layers.31.self_attn.k_norm	256	8,895,078,656
model.language_model.layers.31.self_attn.k_proj	4,194,304	8,899,272,960
model.language_model.layers.31.self_attn.o_proj	16,777,216	8,916,050,176
model.language_model.layers.31.self_attn.q_norm	256	8,916,050,432
model.language_model.layers.31.self_attn.q_proj	33,554,432	8,949,604,864
model.language_model.layers.31.self_attn.v_proj	4,194,304	8,953,799,168
model.language_model.norm	4,096	8,953,803,264
model.visual.blocks.0.attn.proj	1,328,256	8,955,131,520
model.visual.blocks.0.attn.qkv	3,984,768	8,959,116,288
model.visual.blocks.0.mlp.linear_fc1	4,962,512	8,964,078,800
model.visual.blocks.0.mlp.linear_fc2	4,959,360	8,969,038,160
model.visual.blocks.0.norm1	2,304	8,969,040,464
model.visual.blocks.0.norm2	2,304	8,969,042,768
model.visual.blocks.1.attn.proj	1,328,256	8,970,371,024
model.visual.blocks.1.attn.qkv	3,984,768	8,974,355,792
model.visual.blocks.1.mlp.linear_fc1	4,962,512	8,979,318,304
model.visual.blocks.1.mlp.linear_fc2	4,959,360	8,984,277,664
model.visual.blocks.1.norm1	2,304	8,984,279,968
model.visual.blocks.1.norm2	2,304	8,984,282,272
model.visual.blocks.2.attn.proj	1,328,256	8,985,610,528
model.visual.blocks.2.attn.qkv	3,984,768	8,989,595,296
model.visual.blocks.2.mlp.linear_fc1	4,962,512	8,994,557,808
model.visual.blocks.2.mlp.linear_fc2	4,959,360	8,999,517,168
model.visual.blocks.2.norm1	2,304	8,999,519,472
model.visual.blocks.2.norm2	2,304	8,999,521,776
model.visual.blocks.3.attn.proj	1,328,256	9,000,850,032
model.visual.blocks.3.attn.qkv	3,984,768	9,004,834,800
model.visual.blocks.3.mlp.linear_fc1	4,962,512	9,009,797,312
model.visual.blocks.3.mlp.linear_fc2	4,959,360	9,014,756,672
model.visual.blocks.3.norm1	2,304	9,014,758,976
model.visual.blocks.3.norm2	2,304	9,014,761,280
model.visual.blocks.4.attn.proj	1,328,256	9,016,089,536

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Module	Params	Running total
model.visual.blocks.4.attn.qkv	3,984,768	9,020,074,304
model.visual.blocks.4.mlp.linear_fc1	4,962,512	9,025,036,816
model.visual.blocks.4.mlp.linear_fc2	4,959,360	9,029,996,176
model.visual.blocks.4.norm1	2,304	9,029,998,480
model.visual.blocks.4.norm2	2,304	9,030,000,784
model.visual.blocks.5.attn.proj	1,328,256	9,031,329,040
model.visual.blocks.5.attn.qkv	3,984,768	9,035,313,808
model.visual.blocks.5.mlp.linear_fc1	4,962,512	9,040,276,320
model.visual.blocks.5.mlp.linear_fc2	4,959,360	9,045,235,680
model.visual.blocks.5.norm1	2,304	9,045,237,984
model.visual.blocks.5.norm2	2,304	9,045,240,288
model.visual.blocks.6.attn.proj	1,328,256	9,046,568,544
model.visual.blocks.6.attn.qkv	3,984,768	9,050,553,312
model.visual.blocks.6.mlp.linear_fc1	4,962,512	9,055,515,824
model.visual.blocks.6.mlp.linear_fc2	4,959,360	9,060,475,184
model.visual.blocks.6.norm1	2,304	9,060,477,488
model.visual.blocks.6.norm2	2,304	9,060,479,792
model.visual.blocks.7.attn.proj	1,328,256	9,061,808,048
model.visual.blocks.7.attn.qkv	3,984,768	9,065,792,816
model.visual.blocks.7.mlp.linear_fc1	4,962,512	9,070,755,328
model.visual.blocks.7.mlp.linear_fc2	4,959,360	9,075,714,688
model.visual.blocks.7.norm1	2,304	9,075,716,992
model.visual.blocks.7.norm2	2,304	9,075,719,296
model.visual.blocks.8.attn.proj	1,328,256	9,077,047,552
model.visual.blocks.8.attn.qkv	3,984,768	9,081,032,320
model.visual.blocks.8.mlp.linear_fc1	4,962,512	9,085,994,832
model.visual.blocks.8.mlp.linear_fc2	4,959,360	9,090,954,192
model.visual.blocks.8.norm1	2,304	9,090,956,496
model.visual.blocks.8.norm2	2,304	9,090,958,800
model.visual.blocks.9.attn.proj	1,328,256	9,092,287,056
model.visual.blocks.9.attn.qkv	3,984,768	9,096,271,824
model.visual.blocks.9.mlp.linear_fc1	4,962,512	9,101,234,336
model.visual.blocks.9.mlp.linear_fc2	4,959,360	9,106,193,696
model.visual.blocks.9.norm1	2,304	9,106,196,000
model.visual.blocks.9.norm2	2,304	9,106,198,304
model.visual.blocks.10.attn.proj	1,328,256	9,107,526,560
model.visual.blocks.10.attn.qkv	3,984,768	9,111,511,328
model.visual.blocks.10.mlp.linear_fc1	4,962,512	9,116,473,840
model.visual.blocks.10.mlp.linear_fc2	4,959,360	9,121,433,200
model.visual.blocks.10.norm1	2,304	9,121,435,504
model.visual.blocks.10.norm2	2,304	9,121,437,808
model.visual.blocks.11.attn.proj	1,328,256	9,122,766,064
model.visual.blocks.11.attn.qkv	3,984,768	9,126,750,832
model.visual.blocks.11.mlp.linear_fc1	4,962,512	9,131,713,344
model.visual.blocks.11.mlp.linear_fc2	4,959,360	9,136,672,704
model.visual.blocks.11.norm1	2,304	9,136,675,008
model.visual.blocks.11.norm2	2,304	9,136,677,312
model.visual.blocks.12.attn.proj	1,328,256	9,138,005,568
model.visual.blocks.12.attn.qkv	3,984,768	9,141,990,336
model.visual.blocks.12.mlp.linear_fc1	4,962,512	9,146,952,848
model.visual.blocks.12.mlp.linear_fc2	4,959,360	9,151,912,208
model.visual.blocks.12.norm1	2,304	9,151,914,512
model.visual.blocks.12.norm2	2,304	9,151,916,816
model.visual.blocks.13.attn.proj	1,328,256	9,153,245,072
model.visual.blocks.13.attn.qkv	3,984,768	9,157,229,840
model.visual.blocks.13.mlp.linear_fc1	4,962,512	9,162,192,352
model.visual.blocks.13.mlp.linear_fc2	4,959,360	9,167,151,712
model.visual.blocks.13.norm1	2,304	9,167,154,016

continued on next page

Module	Params	Running total
model.visual.blocks.13.norm2	2,304	9,167,156,320
model.visual.blocks.14.attn.proj	1,328,256	9,168,484,576
model.visual.blocks.14.attn.qkv	3,984,768	9,172,469,344
model.visual.blocks.14.mlp.linear_fc1	4,962,512	9,177,431,856
model.visual.blocks.14.mlp.linear_fc2	4,959,360	9,182,391,216
model.visual.blocks.14.norm1	2,304	9,182,393,520
model.visual.blocks.14.norm2	2,304	9,182,395,824
model.visual.blocks.15.attn.proj	1,328,256	9,183,724,080
model.visual.blocks.15.attn.qkv	3,984,768	9,187,708,848
model.visual.blocks.15.mlp.linear_fc1	4,962,512	9,192,671,360
model.visual.blocks.15.mlp.linear_fc2	4,959,360	9,197,630,720
model.visual.blocks.15.norm1	2,304	9,197,633,024
model.visual.blocks.15.norm2	2,304	9,197,635,328
model.visual.blocks.16.attn.proj	1,328,256	9,198,963,584
model.visual.blocks.16.attn.qkv	3,984,768	9,202,948,352
model.visual.blocks.16.mlp.linear_fc1	4,962,512	9,207,910,864
model.visual.blocks.16.mlp.linear_fc2	4,959,360	9,212,870,224
model.visual.blocks.16.norm1	2,304	9,212,872,528
model.visual.blocks.16.norm2	2,304	9,212,874,832
model.visual.blocks.17.attn.proj	1,328,256	9,214,203,088
model.visual.blocks.17.attn.qkv	3,984,768	9,218,187,856
model.visual.blocks.17.mlp.linear_fc1	4,962,512	9,223,150,368
model.visual.blocks.17.mlp.linear_fc2	4,959,360	9,228,109,728
model.visual.blocks.17.norm1	2,304	9,228,112,032
model.visual.blocks.17.norm2	2,304	9,228,114,336
model.visual.blocks.18.attn.proj	1,328,256	9,229,442,592
model.visual.blocks.18.attn.qkv	3,984,768	9,233,427,360
model.visual.blocks.18.mlp.linear_fc1	4,962,512	9,238,389,872
model.visual.blocks.18.mlp.linear_fc2	4,959,360	9,243,349,232
model.visual.blocks.18.norm1	2,304	9,243,351,536
model.visual.blocks.18.norm2	2,304	9,243,353,840
model.visual.blocks.19.attn.proj	1,328,256	9,244,682,096
model.visual.blocks.19.attn.qkv	3,984,768	9,248,666,864
model.visual.blocks.19.mlp.linear_fc1	4,962,512	9,253,629,376
model.visual.blocks.19.mlp.linear_fc2	4,959,360	9,258,588,736
model.visual.blocks.19.norm1	2,304	9,258,591,040
model.visual.blocks.19.norm2	2,304	9,258,593,344
model.visual.blocks.20.attn.proj	1,328,256	9,259,921,600
model.visual.blocks.20.attn.qkv	3,984,768	9,263,906,368
model.visual.blocks.20.mlp.linear_fc1	4,962,512	9,268,868,880
model.visual.blocks.20.mlp.linear_fc2	4,959,360	9,273,828,240
model.visual.blocks.20.norm1	2,304	9,273,830,544
model.visual.blocks.20.norm2	2,304	9,273,832,848
model.visual.blocks.21.attn.proj	1,328,256	9,275,161,104
model.visual.blocks.21.attn.qkv	3,984,768	9,279,145,872
model.visual.blocks.21.mlp.linear_fc1	4,962,512	9,284,108,384
model.visual.blocks.21.mlp.linear_fc2	4,959,360	9,289,067,744
model.visual.blocks.21.norm1	2,304	9,289,070,048
model.visual.blocks.21.norm2	2,304	9,289,072,352
model.visual.blocks.22.attn.proj	1,328,256	9,290,400,608
model.visual.blocks.22.attn.qkv	3,984,768	9,294,385,376
model.visual.blocks.22.mlp.linear_fc1	4,962,512	9,299,347,888
model.visual.blocks.22.mlp.linear_fc2	4,959,360	9,304,307,248
model.visual.blocks.22.norm1	2,304	9,304,309,552
model.visual.blocks.22.norm2	2,304	9,304,311,856
model.visual.blocks.23.attn.proj	1,328,256	9,305,640,112
model.visual.blocks.23.attn.qkv	3,984,768	9,309,624,880
model.visual.blocks.23.mlp.linear_fc1	4,962,512	9,314,587,392

continued on next page

Module	Params	Running total
model.visual.blocks.23.mlp.linear_fc2	4,959,360	9,319,546,752
model.visual.blocks.23.norm1	2,304	9,319,549,056
model.visual.blocks.23.norm2	2,304	9,319,551,360
model.visual.blocks.24.attn.proj	1,328,256	9,320,879,616
model.visual.blocks.24.attn.qkv	3,984,768	9,324,864,384
model.visual.blocks.24.mlp.linear_fc1	4,962,512	9,329,826,896
model.visual.blocks.24.mlp.linear_fc2	4,959,360	9,334,786,256
model.visual.blocks.24.norm1	2,304	9,334,788,560
model.visual.blocks.24.norm2	2,304	9,334,790,864
model.visual.blocks.25.attn.proj	1,328,256	9,336,119,120
model.visual.blocks.25.attn.qkv	3,984,768	9,340,103,888
model.visual.blocks.25.mlp.linear_fc1	4,962,512	9,345,066,400
model.visual.blocks.25.mlp.linear_fc2	4,959,360	9,350,025,760
model.visual.blocks.25.norm1	2,304	9,350,028,064
model.visual.blocks.25.norm2	2,304	9,350,030,368
model.visual.blocks.26.attn.proj	1,328,256	9,351,358,624
model.visual.blocks.26.attn.qkv	3,984,768	9,355,343,392
model.visual.blocks.26.mlp.linear_fc1	4,962,512	9,360,305,904
model.visual.blocks.26.mlp.linear_fc2	4,959,360	9,365,265,264
model.visual.blocks.26.norm1	2,304	9,365,267,568
model.visual.blocks.26.norm2	2,304	9,365,269,872
model.visual.merger.linear_fc1	21,238,272	9,386,508,144
model.visual.merger.linear_fc2	18,878,464	9,405,386,608
model.visual.merger.norm	2,304	9,405,388,912
model.visual.patch_embed.proj	1,770,624	9,407,159,536
model.visual.pos_embed	2,654,208	9,409,813,744